

Optimization and Performance Analysis of Resource Allocation in Quantum-Centric Supercomputing Environments

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Abstract

In this paper, we consider the problem of determining the optimal allocation of resource capacities in quantum-centric supercomputing environments in which, unlike traditional high-performance computing environments, the system architectures comprise combinations of quantum and classical processing units and the application workloads comprise computation patterns spanning quantum and classical processing units. We derive a general theoretical foundation for the mathematical optimization and performance analysis of resource allocation in such quantum-centric supercomputing environments, including a mathematical analysis of basic computational phases at the core of application workload patterns and a mathematical analysis of fairly common reservation-based resource allocation policies. Then we derive an efficient numerical solution of the general resource capacity allocation optimization problem which yields an optimal resource allocation policy, provably establishing theoretical properties on the guaranteed convergence of the corresponding iterative process to a unique fixed point and providing fundamental insights into resource allocation in quantum-centric supercomputing environments. Numerical experiments demonstrate that our optimal allocation policy provides significant performance benefits over the existing reservation-based allocation policy across different application workloads. This also includes evaluating the performance of the basic core computational phases across different workloads and demonstrating the very high accuracy of our closed-form approximations for the corresponding fork-join queueing networks, which are of interest in their own right.

1 Introduction

The evolution and growth of quantum computing has been tremendous since its origins, starting with the original proposal by Benioff in 1980 for a quantum version of a Turing machine based on the equations of quantum mechanics [10]. This was shortly followed by Feynman’s keynote at the First Conference on the Physics of Computation, jointly organized in 1981 by IBM and MIT, in which he proposed a “quantum simulator” that, unlike a universal Turing machine, specializes in simulating quantum mechanical phenomena [17].

Feynman closed this keynote with his now-famous remark: “. . . nature isn’t classical, dammit, and if you want to make a simulation of nature, you’d better make it quantum mechanical, and by golly it’s a wonderful problem, because it doesn’t look so easy.”

Since then, quantum computing has increasingly captured broad attention across various fields of science, engineering, and business due to its potential to solve certain problems much faster than classical computers. Over the past few years, there have been immense advances in quantum computing hardware. This includes quantum computers that use logical quantum gates to execute algorithms, such as those based on superconducting quantum bits (e.g., IBM, Google, Rigetti), trapped-ion systems (e.g., IonQ, Quantinuum), cold-atom devices (e.g., QuEra), and photonic devices (e.g., Xanadu, PsiQuantum). These quantum computers are capable in principle of executing any algorithm in the complexity class for quantum polynomial-time computation, known as BQP.

In combination with such progress and growth in quantum computing hardware, there have been fundamental advances in quantum algorithms across different computational problem domains that offer tremendous potential for truly significant performance advantages in the execution of such algorithms on quantum computers over their classical counterparts for certain problems. This includes examples in factoring large numbers [63, 64], unstructured search [22, 23], parameter estimation [11], linear systems of equations [25], Monte Carlo methods [32, 43], finance [54, 72], Bayesian experimental design [21], and optimization [1, 26]. Recent examples have even demonstrated the benefit of quantum computers over brute-force classical approaches in simulating quantum systems [30, 58]. These advances and trends in both quantum computing hardware and algorithms have helped to bring about major paradigm shifts in the evolution of high-performance supercomputing environments. Such paradigm shifts include foundational changes in the core designs of high-performance computing (HPC) system infrastructures that integrate both quantum and classical computing through a combination of quantum processing units (QPUs), graphical processing units (GPUs), and central processing units (CPUs) [9, 39, 60, 62, 65, 69]. In fact, Seelam et al. [60] have even defined the first reference architecture for so-called quantum-centric supercomputing (QCSC) environments. These paradigm shifts also include foundational changes in the core designs of HPC application workloads that combine different patterns of computation across the various QPUs, GPUs, and CPUs. This results in application workloads comprising jobs that require only classical GPUs and CPUs, only quantum QPUs, or a mixture of classical GPUs and CPUs and quantum QPUs. At the same time, although there have been considerable research studies of resource allocation under application workloads based on GPUs and CPUs, our theoretical understanding of the mathematical optimization and performance analysis of resource capacity allocation in high-performance supercomputing environments is far more limited and largely unexplored under QPU-only and QPU-CPU-combined workloads, especially with respect to the latter.

In this paper, we seek to gain such a theoretical understanding of the foundations for the mathematical optimization and performance analysis of resource capacity allocation in QCSC environments comprising a combination of QPUs, GPUs and CPUs. A key design issue concerns the patterns of application workloads in these QCSC environments that com-

prise various combinations of core parallel and sequential computational phases across the different processing units. In particular, these two basic core components of QCSC application workloads consist of: (1) quantum and classical computational tasks that are executed simultaneously in parallel on both QPUs and GPUs; and (2) quantum and classical computational tasks that are executed individually in sequence on QPUs and GPUs in any order. The former concern fork-join queueing networks, while the latter concerns tandem queueing networks, where both often involve heterogeneous service times. Hence, another key design issue concerns the performance of these core parallel and sequential computational phases as the building blocks of QCSC application workloads. Yet another key design issue concerns the resource allocation policies that should be employed in QCSC environments. While it is not uncommon for reservation-based resource allocation policies to be deployed in these QCSC environments, the optimal resource capacity allocation policy is not well understood nor is the performance of such an optimal policy relative to the existing reservation-based policies.

The allocation of resources in stochastic networks comprising fork-join queueing structures or tandem queueing structures has received considerable attention in the research literature. In particular, studies of resource allocation in fork-join queueing networks have arisen within the context of warehouse-scale computers [16]; user-facing datacenters [52]; distributed, telecommunication, and storage systems [40]; and the tradeoffs between parallel execution and synchronization penalties [29, 38]. Analogously, studies of resource allocation in various combinations of tandem queues have arisen in many different contexts, such as communication networks, manufacturing, power grids, and supply chains [41]. Our study of QCSC environments considers tailored mixtures of fork-join queueing networks and tandem queueing networks in complex combinations with servers that represent the highly diverse characteristics of QPUs, GPUs, and CPUs. Moreover, our study seeks to derive a general theoretical foundation that provides a better understanding of optimal resource capacity allocation in such QCSC environments.

Our Contributions. In this paper, we establish a general theoretical foundation for the mathematical optimization and performance analysis of resource capacity allocation in QCSC environments. To the best of our knowledge, this is the first such study which leads to the following important technical contributions. First, we formulate mathematical models of the application workload patterns in these QCSC environments, with an emphasis on jobs requiring combinations of QPUs, GPUs and CPUs, and then focus on a particular instance that spans the combination of parallel and sequential computational phases at the core of these patterns. Second, we derive a mathematical analysis of the core parallel and sequential computational phases of QCSC application workloads, which includes exact closed-form solutions for the expected sojourn time of the sequential computational phase and highly accurate closed-form approximations for the expected sojourn time of the parallel computational phase. The latter theoretical results provide the first closed-form expressions for the expected sojourn time of the corresponding fork-join queueing network with non-homogeneous service rates, and thus are of interest in their own right. Third, we derive a matrix-analytic analysis of the performance of QCSC environments under reservation-based

resource allocation policies where the system resources are reserved by jobs in the order of their arrival. Since it is not uncommon to find this approach employed in practice, this analysis is included for comparison against an optimal resource capacity allocation policy. Fourth, we formulate a general resource capacity allocation optimization problem in the context of QCSC environments and derive an efficient numerical solution for the optimal capacity allocation policy that differs significantly from the reservation-based resource allocation approach. Our optimization problem formulation addresses the fundamental tradeoff between the performance experienced by jobs submitted to the QCSC environment and the financial measures encountered by the host of the environment. Finally, we conduct a large collection of numerical experiments to assess various aspects of the theoretical foundations established in our study. This includes evaluating the performance of the basic core parallel and sequential computational phases of QCSC application workloads as a function of various system and workload parameters. These numerical results further demonstrate the very high accuracy of our closed-form approximations for the fork-join queueing networks of the core parallel computational phases. Our numerical experiments also illustrate the properties of our optimal resource capacity allocation policy as a function of various system and workload parameters, as well as demonstrating that the optimal capacity allocation policy provides significant performance benefits over existing reservation-based allocation policies across different application workloads.

Organization of paper. Section 2 presents our mathematical models of QCSC system environments and application workloads, followed by a mathematical analysis of the two basic core components of QCSC application workloads. We then derive in Section 4 a matrix-analytic analysis of the QCSC environment under a reservation-based resource allocation policy not uncommonly employed in practice, followed by the derivation of a mathematical framework for determining the optimal resource capacity allocation policy. A representative sample of our collection of numerical experiments is presented in Section 6, followed by concluding remarks. Due to space constraints, the proofs of our main theoretical results are presented in Appendix A, with subsequent appendices providing additional technical details.

2 Mathematical Models

The applications of QCSC environments are generally characterized by a directed acyclic graph. Here, the directed edges from the set of parent nodes to a child node in the graph define the set of tasks associated with the parent nodes that must be completed before the task associated with the child node can start execution. Based on existing and anticipated application patterns [9,20,62], the basic core components of these directed acyclic graphs that make up QCSC application workloads are represented by: (1) a parallel computational phase comprising a single fork-join pair of quantum and classical computational tasks executed in parallel; and (2) a sequential computational phase comprising a single pair of quantum and classical computational tasks executed in series in either order. Then the QCSC application workloads consist of a mixture of these two basic core components, where quantum compu-

tational tasks are executed on system QPUs and classical computational tasks are executed on system GPUs. As a representative model of these workloads, we primarily consider two classes of QCSC applications with different structures for the computational phases depicted in the directed acyclic graph. The first class, called *parallel-sequential applications*, consists of a parallel computational phase followed by a sequential computational phase. The second class, called *sequential-parallel applications*, consists of a sequential computational phase followed by a parallel computational phase. Although not a strict requirement, we envision that both application classes start with a classical initialization task executed on system CPUs and end with a classical termination task executed on system CPUs.

Jobs arrive to the QCSC environment according to a Poisson process with rate λ . With probability $p_{\parallel}$, the arriving job belongs to the parallel-sequential application class, and with probability $1 - p_{\parallel}$, the arriving job belongs to the sequential-parallel application class. In both application classes, the sequential computational phase consists of a quantum computational task followed by a classical computational task with probability $p_{\circ}$, and consists of a classical computational task followed by a quantum computational task with probability $1 - p_{\circ}$. The quantum computational tasks of any parallel computational phase or sequential computational phase can involve parallel quantum computations requiring execution on $M_Q \geq 1$ QPUs. Similarly, the classical computational tasks of any parallel computational phase or sequential computational phase can involve parallel classical computations requiring execution on $M_G \geq 1$ GPUs. As our starting point, we shall consider $M_Q = 1$ for all parallel computational phases and $M_G = 1$ for all sequential computational phases. While this helps to reduce the complexity of the mathematical analysis and optimization problems at hand, it is important to note that queueing-theoretic performance metrics such as the expected sojourn time in a system with M_Q QPUs operating at unit speed converges asymptotically to the corresponding queueing-theoretic performance metrics in our assumed single QPU with speed M_Q in the limit as the system utilization $\rho_Q \rightarrow 1$; and similarly for a system with M_G GPUs operating at unit speed converging asymptotically to our assumed single GPU with speed M_G in the limit as the system utilization $\rho_G \rightarrow 1$ [14, 71].

For the parallel computational phase of the parallel-sequential application class, the service times of quantum computational tasks on system QPUs follow an exponential distribution with rate μ_Q^{PP} , and the service times of classical computational tasks on system GPUs follow an exponential distribution with rate μ_G^{PP} . Similarly, for the parallel computational phase of the sequential-parallel application class, the service times of quantum computational tasks on system QPUs follow an exponential distribution with rate μ_Q^{SP} , and the service times of classical computational tasks on system GPUs follow an exponential distribution with rate μ_G^{SP} . Analogously, for the sequential computational phase of the parallel-sequential application class, the service times of quantum computational tasks on system QPUs follow an exponential distribution with rate $\mu_Q^{PS_q}$ or $\mu_Q^{PS_g}$, depending on whether the order of the series starts with a quantum (S_q) or classical (S_g) task, and the service times of classical computational tasks on system GPUs follow an exponential distribution with rate $\mu_G^{PS_q}$ or $\mu_G^{PS_g}$, depending on whether the series order starts with a quantum (S_q) or classical (S_g) task. Similarly, for the sequential computational phase of the sequential-parallel application class,

the service times of quantum computational tasks on system QPUs follow an exponential distribution with rate $\mu_Q^{SS_q}$ or $\mu_Q^{SS_g}$, depending on whether the order of the series starts with a quantum (S_q) or classical (S_g) task, and the service times of classical computational tasks on system GPUs follow an exponential distribution with rate $\mu_G^{SS_q}$ or $\mu_G^{SS_g}$, depending on whether the series order starts with a quantum (S_q) or classical (S_g) task.

The service times for a classical initialization and termination task on system CPUs follow an exponential distribution with rate μ_I^P, μ_I^S and μ_T^P, μ_T^S , respectively, such that $\mu_I^P, \mu_I^S, \mu_T^P, \mu_T^S \gg \mu_Q^{PP}, \mu_Q^{PS_q}, \mu_Q^{PS_g}, \mu_Q^{SS_q}, \mu_Q^{SS_g}, \mu_Q^{SP}, \mu_G^{PP}, \mu_G^{PS_q}, \mu_G^{PS_g}, \mu_G^{SS_q}, \mu_G^{SS_g}, \mu_G^{SP}$. There are three different instances of this QCSC application workload model. The first, called *balanced application workloads*, consists of homogeneous service rates where $\mu_Q^\ell = \mu_G^\ell$ for $\ell \in \{PP, PS_q, PS_g, SS_q, SS_g, SP\}$; the second instance, called *quantum-dominant application workloads*, consists of nonhomogeneous service rates where $\mu_Q^\ell < \mu_G^\ell$ for $\ell \in \{PP, PS_q, PS_g, SS_q, SS_g, SP\}$; and the third instance, called *classical-dominant application workloads*, consists of nonhomogeneous service rates where $\mu_Q^\ell > \mu_G^\ell$ for $\ell \in \{PP, PS_q, PS_g, SS_q, SS_g, SP\}$. Although all three application workload models are considered in our study, it is important to point out that existing and anticipated application patterns suggest that balanced workloads are less likely to be found in QCSC environments in practice and it is strongly believed that this trend will persist [62]. Quantum-dominant and classical-dominant workloads are expected to be far more prevalent, and thus these two workloads represent an important emphasis in our study.

While our assumptions of Poisson job arrivals and exponential service times help to reduce the complexity of the mathematical analysis and optimization problems at hand, it is critically important to note that there is little and limited data available on the arrival and service patterns for QCSC environments, and whatever data might be available today is highly unlikely to be representative of future QCSC application workloads [37]. Similarly, it is critically important to note that there is little and limited data available on the number of QPUs and GPUs required by applications in QCSC environments, and whatever data might be available today is highly unlikely to be representative of future QCSC application workloads, which is another important reason for our focus on $M_Q = 1$ and $M_G = 1$. The field is evolving so quickly with very little in the way of a fundamental analysis of performance in QCSC environments, making it premature to know the actual distributions of arrival and service patterns and the actual number of required QPUs and GPUs in future QCSC application workloads. Meanwhile, there is an important gap in our understanding of fundamental issues related to the design, performance and optimization of QCSC environments. Given such few studies of this type in the field, we seek to address this gap in the research literature with our study providing the first mathematical analysis and optimization of fundamental performance and related theoretical issues in QCSC environments, to the best of our knowledge.

3 Mathematical Analysis of Core Computational Phases

The two basic core components of QCSC application workloads consist of either a sequential computational phase or a parallel computational phase. We consider in turn a mathematical analysis of each of these two basic core components of QCSC application workloads in isolation, first deriving a closed-form expressions for the expected sojourn time of the sequential computational phase and then deriving a closed-form expressions for the expected sojourn time of the parallel computational phase.

3.1 Sequential Computational Phase

The stochastic network corresponding to the sequential computational phase of QCSC application workloads in isolation represents an open Jackson queueing network [7, 27]. Such jobs arrive according to a Poisson process with rate γ , each of which is routed with probability p_o to single-server queue 1 for a system QPU and is routed with probability $1 - p_o$ to single-server queue 2 for a system GPU. Upon completion of the quantum computational task at server 1, the corresponding jobs are routed directly to single-server queue 3 for a system GPU; and upon completion of the classical computational task at server 2, the corresponding jobs are routed directly to single-server queue 4 for a system QPU. The arrivals to single-server queues i follow Poisson processes with rates $\gamma_1 = \gamma_3 = p_o\gamma$ and $\gamma_2 = \gamma_4 = (1 - p_o)\gamma$; and the service times for single-server queues i follow exponential distributions with rates $\hat{\mu}_i$, $i \in \{1, \dots, 4\}$, where $\hat{\mu}_1 = \mu_Q^{PS}$, $\hat{\mu}_2 = \mu_G^{SP}$, $\hat{\mu}_3 = \mu_G^{PS}$, $\hat{\mu}_4 = \mu_Q^{SP}$. Then, jobs depart the system upon completion of their computational tasks at server 3 or server 4. From Burke's Theorem [13] and the quasireversibility of such stochastic networks [28, 44], we also have that the departure process from single-server queue 3 and single-server queue 4 are Poisson with rate $p_o\gamma$ and $(1 - p_o)\gamma$, respectively.

Stochastic networks of this form have received considerable attention in the research literature; see, e.g., [3, 7, 12, 27, 28, 35]. It is generally well known that these open Jackson queueing networks have a product-form solution, which implies that each single-server queue of the stochastic network can be analyzed in a separable manner from the rest of the network. We therefore have that the expected sojourn time at server i is given by

$$\mathbb{E}[T_{PF}^{(i)}] = \frac{1}{\hat{\mu}_i - \gamma_i}, \quad i \in \{1, \dots, 4\}, \quad (1)$$

and that the expected overall sojourn time of the sequential computational phase of QCSC application workloads in isolation is obtained as the sum of the expected sojourn time at each of the servers i above; namely, we have

$$\mathbb{E}[T_{PF}] = \frac{1}{\hat{\mu}_1 - \gamma_1} + \frac{1}{\hat{\mu}_2 - \gamma_2} + \frac{1}{\hat{\mu}_3 - \gamma_3} + \frac{1}{\hat{\mu}_4 - \gamma_4}. \quad (2)$$

Although we primarily focus on a single pair of alternating quantum and classical computational tasks executed in series in either order, the corresponding results for a general number of sequential computational phases would be obtained in a similar manner.

3.2 Parallel Computational Phase

The stochastic network corresponding to the parallel computational phase of QCSC application workloads in isolation represents a fork-join queueing network with two parallel servers [18, 19]. Such jobs arrive according to a Poisson process with rate γ , each of which forks into a pair of quantum and classical computational tasks to be executed in parallel with one routed to a single-server queue for the system QPU and the other routed to a single-server queue for the system GPU. The service times for quantum computational tasks at the system QPU server and for classical computational tasks at the system GPU server each follow an exponential distribution with rate μ_Q^{PP} and μ_G^{PP} , respectively. Without loss of generality, let $\hat{\mu}_1 = \min\{\mu_Q^{PP}, \mu_G^{PP}\}$ and $\hat{\mu}_2 = \max\{\mu_Q^{PP}, \mu_G^{PP}\}$, and thus $\hat{\mu}_1 \leq \hat{\mu}_2$. Then, jobs depart the system only upon completion of their computational tasks at both server 1 and server 2. Define the critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$. For the purpose of analyzing quantum-dominant and classical-dominant application workloads relative to balanced application workloads, define the heterogeneity ratio $r := \hat{\mu}_2/\hat{\mu}_1 \in [1, \infty)$.

Stochastic networks of this form have received considerable attention in the research literature; see, e.g., [6, 18, 19, 42, 46, 53, 59, 66]. In the case of balanced application workloads with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$, it is generally well known that the expected sojourn time of jobs in these fork-join queueing networks is given by

$$\mathbb{E}[T_{FJ}] = \frac{12 - \gamma/\hat{\mu}}{8(\hat{\mu} - \gamma)}, \quad (3)$$

due to Nelson and Tantawi [46]. Hence, the expected overall sojourn time of the parallel computational phase of QCSC application workloads in isolation can be expressed as in (3).

On the other hand, as discussed in Section 2, quantum-dominant workloads with $\mu_Q^{PP} < \mu_G^{PP}$ and classical-dominant workloads with $\mu_G^{PP} < \mu_Q^{PP}$ are expected to be far more prevalent than balanced workloads in QCSC environments. In strong contrast to (3), the corresponding fork-join queueing network under nonhomogeneous service rates generally represents a difficult queueing-theoretic problem for which there is no known exact expression for the expected sojourn time. To address this queueing-theoretic problem, we derive a few mathematical approximations based on different forms of interpolation. Our first approximation approach consists of deriving upper and lower bounds and then interpolating between these two bounds. Our second approximation approach consists of deriving light-heavy traffic interpolation approximations. A discussion of this approach, developed in large part by Reiman and Simon [55–57], is presented in Appendix B to which we refer for additional technical details. We derive in turn each of our closed-form mathematical approximations and related theoretical results for two-server fork-join queueing networks with nonhomogeneous service rates, which are of interest in their own right, noting that each of our approximations yields the exact result in (3) for fork-join queueing networks with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$. The numerical results in Section 6 demonstrate that both approaches provide very accurate results for the analysis of otherwise intractable fork-join queueing systems.

3.2.1 Upper-Lower Bound Interpolation

Our upper-lower bound interpolation approach is based on a convex combination of an upper and a lower bound on the expected sojourn time in the fork-join queueing network under nonhomogeneous service rates $\hat{\mu}_1 < \hat{\mu}_2$. We start by first deriving the following theoretical result for these upper and lower bounds.

Lemma 1. *The expected sojourn time of the two-server fork-join queueing network $\mathbb{E}[T_{FJ}]$ is upper bounded by*

$$\mathbb{E}[T_{FJ}] \leq \mathbb{E}[T_{FJ}^{UB}] = \frac{1}{\hat{\mu}_1 - \gamma} + \frac{1}{\hat{\mu}_2 - \gamma} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2 - 2\gamma}. \quad (4)$$

Furthermore, the expected sojourn time of the two-server fork-join queueing network is lower bounded by

$$\mathbb{E}[T_{FJ}] \geq \mathbb{E}[T_{FJ}^{LB}] = \frac{1}{\hat{\mu}_1 - \gamma}. \quad (5)$$

Recall $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$ and $r := \hat{\mu}_2/\hat{\mu}_1 \in [1, \infty)$. Observe that, in the case of $r = 1$ with $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$, we have $\mathbb{E}[T_{FJ}^{UB}] = (3/2) \cdot \mathbb{E}[T_{MM1}]$ and $\mathbb{E}[T_{FJ}^{LB}] = \mathbb{E}[T_{MM1}]$ where $\mathbb{E}[T_{MM1}]$ is the expected sojourn time of an M/M/1 queue with arrival rate γ and service rate $\hat{\mu}$, resulting in a 50% gap between the upper and lower bounds with the exact expected sojourn time in (3) falling strictly between the two bounds. Moreover, in the light-traffic limit as $\gamma \rightarrow 0$, $\mathbb{E}[T_{FJ}]$ tends toward $\mathbb{E}[T_{FJ}^{UB}]$ with $\mathbb{E}[T_{FJ}]/\mathbb{E}[T_{FJ}^{UB}] \rightarrow 1$; and, in the heavy-traffic limit as $\gamma \rightarrow \hat{\mu}^*$, $\mathbb{E}[T_{FJ}]$ tends toward $\mathbb{E}[T_{FJ}^{LB}]$ with $\mathbb{E}[T_{FJ}]/\mathbb{E}[T_{FJ}^{LB}] \rightarrow 11/8$. On the other hand, we observe that the gap between the upper and lower bounds decreases with increasing $r > 1$. Moreover, in the limit as $r \rightarrow \infty$, we have that $\mathbb{E}[T_{FJ}^{UB}]$ converges from above to $\mathbb{E}[T_{FJ}^{LB}] = 1/(\hat{\mu}_1 - \gamma)$ and thus the bounds become tight asymptotically with the expected sojourn time well approximated by that of the bottleneck M/M/1 queue.

We now exploit the above results to obtain the desired upper-lower bound interpolation approximation for the expected sojourn time of jobs in the two-server fork-join queueing network of interest under unbalanced application workloads.

Theorem 1. *Consider the sojourn time T_{FJ} of jobs in a fork-join queueing network comprising two parallel servers where arrivals follow a Poisson process with rate γ and service times follow an exponential distribution with nonhomogeneous rates $\hat{\mu}_1$ and $\hat{\mu}_2$ at server 1 and 2, respectively. Without loss of generality, let $\hat{\mu}_1 = \mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. The upper-lower bound interpolation approximation is then given by*

$$\mathbb{E}[T_{FJ}] = (1 - \hat{\alpha}) \cdot \left(\frac{1}{\hat{\mu}_1 - \gamma} + \frac{1}{\hat{\mu}_2 - \gamma} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2 - 2\gamma} \right) + \hat{\alpha} \cdot \left(\frac{1}{\hat{\mu}_1 - \gamma} \right), \quad \gamma \in [0, \hat{\mu}_1), \quad (6)$$

where $\hat{\alpha} = (\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)/8$.

Remark 1. *It is important to note that, upon substituting $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}$ in (6), our upper-lower bound interpolation approximation recovers the exact expected sojourn time of jobs in the two-server fork-join queueing network of interest under balanced application workloads, as specifically expressed in (3).*

3.2.2 Light-Heavy Traffic Interpolation

Our light-heavy traffic interpolation approach is based on a combination of light-traffic derivatives associated with the Taylor series expansion of $\mathbb{E}[T_{FJ}]$ at $\gamma = 0$ and the heavy-traffic limit of $\mathbb{E}[T_{FJ}]$ as γ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$. The light-traffic theory rigorously developed in [57] enable us in principle to calculate the derivative of any order n , thus providing the means to fully determine an exact expression for $\mathbb{E}[T_{FJ}]$ in terms of its Taylor series expansion and the corresponding light-traffic derivatives for all n . However, as is often the case in practice, the calculations of these light-traffic derivatives quickly become extremely involved. We therefore take the approach of combining a few of the light-traffic derivatives with the heavy-traffic limit. In particular, we derive the 0^{th} and 1^{st} order light-traffic derivatives together with an approximation for the heavy-traffic limit, and then combine these results to obtain our 0^{th} and 1^{st} order light-heavy traffic interpolation approximations.

Consider the sojourn time T_{FJ} of a tagged job that arrives at time $t = 0$ whose service times v_1 and v_2 are random variables that respectively follow an exponential distribution with mean $\mathbb{E}[v_1] = \hat{\mu}_1$ and $\mathbb{E}[v_2] = \hat{\mu}_2$, where v_1 and v_2 are independent of all else. The 0^{th} order light-traffic derivative can be obtained by definition as

$$\mathbb{E}[\psi(\emptyset)] := \mathbb{E}[T_{FJ} \mid \text{no arrivals over } (-\infty, 0)], \quad (7)$$

noting that any arrival at time $t \in (0, \infty)$ will be serviced after the tagged job and therefore not impact T_{FJ} . We then have the following theoretical result for $\mathbb{E}[\psi(\emptyset)]$.

Lemma 2. *The 0^{th} order light-traffic derivative of the two-server fork-join queueing network is given by*

$$\mathbb{E}[\psi(\emptyset)] = \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \quad (8)$$

Turning to the 1^{st} order light-traffic derivative, we consider the sojourn time T_{FJ} of a tagged job that arrives at time 0 with exponential service times v_1 and v_2 as above, given that another job arrives at time $t \in (-\infty, \infty)$ whose service times w_1 and w_2 are random variables that respectively follow an exponential distribution with mean $\mathbb{E}[w_1] = \hat{\mu}_1$ and $\mathbb{E}[w_2] = \hat{\mu}_2$, where v_1, v_2, w_1, w_2 are independent of all else. The 1^{st} order light-traffic derivative can be obtained by definition as $\mathbb{E}[\psi(\{t\})] := \mathbb{E}[T_{FJ} \mid \text{exactly one arrival at time } t \text{ over } (-\infty, \infty)]$ and

$$\left. \frac{d\mathbb{E}[T_{FJ}]}{d\gamma} \right|_{\gamma=0} = \int_{-\infty}^{\infty} (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt. \quad (9)$$

We then have the following theoretical result for (9).

Lemma 3. *The 1^{st} order light-traffic derivative of the two-server fork-join queueing network is given by*

$$\left. \frac{d\mathbb{E}[T_{FJ}]}{d\gamma} \right|_{\gamma=0} = \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4}. \quad (10)$$

Recall $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\} = \hat{\mu}_1$ and $r := \hat{\mu}_2/\hat{\mu}_1 \in [1, \infty)$. Let us next consider the heavy-traffic limit of the expected sojourn time $\mathbb{E}[T_{FJ}]$ of jobs in the two-server fork-join queueing network under any given $\hat{\mu}_1$ and $\hat{\mu}_2$ as γ approaches its critical value μ^* , namely

$$\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma) \mathbb{E}[T_{FJ}] = h^*. \quad (11)$$

Our approach consists of first establishing upper and lower bounds on the heavy-traffic limit h^* as $r \rightarrow 1$ and as $r \rightarrow \infty$, and then obtaining an upper-lower bound interpolation approximation for h^* . The corresponding theoretical results are presented as follows.

Lemma 4. *The heavy-traffic limit of the two-server fork-join queueing network is bounded above and below by $\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma) \mathbb{E}[T_{FJ}] = h^* \in [1, 11/8]$. Furthermore, the upper-lower bound interpolation approximation for this heavy-traffic limit is given by*

$$h^* \approx h(r) = 1 + \frac{3}{8} r^{-\beta} \in [1, 11/8], \quad r \in [1, \infty), \quad \beta \in (0, \infty). \quad (12)$$

The upper-lower bound interpolation approximation $h(r)$ decays more rapidly under larger control parameters β for a given heterogeneity ratio r . Extensive simulation results strongly suggest that the transition from $h(r)$ around the upper bound of $11/8$ toward the lower bound of 1 for a given $r \in (1, \infty)$ is very sharp. For example, the heavy-traffic limit $h(r)$ has already decayed to approximately 1.008 for $r = 1.5$. Matching this estimate of $h(1.5) \approx 1.008$ to (12) yields $\beta = \log(0.375/0.008)/\log(1.5) \approx 9.5$.

We now exploit the above results to obtain the desired two light-heavy traffic interpolation approximations $\mathbb{E}[T_{FJ}^{(0)}]$ and $\mathbb{E}[T_{FJ}^{(1)}]$ for the expected sojourn time of jobs in the two-server fork-join queueing network of interest under unbalanced application workloads.

Theorem 2. *Consider the sojourn time T_{FJ} of jobs in a fork-join queueing network comprising two parallel servers where arrivals follow a Poisson process with rate γ and service times follow an exponential distribution with nonhomogeneous rates $\hat{\mu}_1$ and $\hat{\mu}_2$ at server 1 and 2, respectively. Further let $f(\gamma) = \mathbb{E}[T_{FJ}]$ and, without loss of generality, $\hat{\mu}_1 = \mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. The 0^{th} and 1^{st} order light-heavy traffic interpolation approximations $\mathbb{E}[T_{FJ}^{(0)}] = \tilde{f}_0(\gamma)$ and $\mathbb{E}[T_{FJ}^{(1)}] = \tilde{f}_1(\gamma)$ for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1)$, are then given by*

$$\mathbb{E}[T_{FJ}^{(0)}] = \tilde{f}_0(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \frac{h(r)}{\hat{\mu}_1} + \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)}, \quad (13)$$

$$\mathbb{E}[T_{FJ}^{(1)}] = \tilde{f}_1(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2 \hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)}. \quad (14)$$

Remark 2. *It is important to note that, upon substituting $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}$ in (13) and (14), our light-heavy traffic interpolation approximations recover the exact expected sojourn time of jobs in the two-server fork-join queueing network of interest under balanced application workloads, as specifically expressed in (3).*

4 Mathematical Analysis of Reserved Allocation

We now consider our QCSC application and system models of Section 2 under a reservation-based resource allocation policy in which the resources of the QCSC environment are reserved by jobs in the order of their arrival. This is not an uncommon approach in practice for many QCSC environments. Recall that each arriving job belongs to the parallel-sequential application class with probability $p_{\parallel}$ and belongs to the sequential-parallel application class with probability $1 - p_{\parallel}$, where the former comprises a parallel computational phase followed by a sequential computational phase and the latter comprises a sequential computational phase followed by a parallel computational phase. The sequential computational phase of both application classes consists of a quantum computational task followed by a sequential computational task with probability p_o , and consists of a sequential computational task followed by a quantum computational task with probability $1 - p_o$. An arriving job of each application class starts with a classical initialization task and ends with a classical termination task. Due to the reservation-based policy of this section, each job reserves the system QPUs, GPUs and CPUs throughout its execution without any overlap with other jobs that wait in a queue for the entire QCSC system.

More formally, consider a continuous-time Markov process $\{X(t); t \in \mathbb{R}_+\}$ on the state space $\hat{\Omega} = \bigcup_{i=0}^{\infty} \hat{\Omega}_i$ where $\hat{\Omega}_0 := \{0\}$ and $\hat{\Omega}_i := \{(i, j_1, j_2, k, q, g) \mid i \in \mathbb{Z}_+, j_1 \in \{P, S\}, j_2 \in \{P, S, S_{gq}, S_{qq}\}, k \in \{I, C, T\}, q \in \{0, 1\}, g \in \{0, 1\}\}$, for the nonnegative reals $\mathbb{R}_+ := \mathbb{R}_{\geq 0}$ with $\mathbb{R}_{\geq k} := [k, \infty)$, and nonnegative integers $\mathbb{Z}_+ := \mathbb{Z}_{\geq 0}$ with $\mathbb{Z}_{\geq k} := \{k, k+1, k+2, \dots\}$. Here, the state-vector variable i denotes the total number of jobs in the QCSC environment, j_1 denotes whether the job in service belongs to the parallel-sequential application class (P) or the sequential-parallel application class (S), j_2 denotes whether the application in service is executing a parallel computational phase (P) or a sequential computational phase (S) that comprises a classical computational task followed by a quantum computational task (S_{gq}) or a quantum computational task followed by a classical computational task (S_{qq}), k denotes whether the job in service is currently executing its initialization phase (I) or termination phase (T) on the system CPUs or executing its quantum/classical computational phases (C), q denotes whether or not the job in service is currently executing a quantum computational task on the system QPUs, and g denotes whether or not the job in service is currently executing a classical computational task on the system GPUs. We shall refer to the set $\hat{\Omega}_i$ as level i .

Define $[M] := \{1, \dots, M\}$, $M < \infty$. Let $x_{i,m} \in \hat{\Omega}_i$ be a pseudo-lexicographic ordering of the elements of $\hat{\Omega}_i$, $i \in \mathbb{Z}_{\geq 1}$, $m \in [M]$; similarly, let $\hat{\Omega}_0 = \{x_0\}$. The ordering $x_{i,m}$ for level $i \in \mathbb{Z}_{\geq 1}$ is then given by: $(i, P, P, I, 0, 0)$; $(i, P, P, C, 0, 1)$; $(i, P, P, C, 1, 0)$; $(i, P, P, C, 1, 1)$; $(i, P, S_{gq}, C, 0, 1)$; $(i, P, S_{gq}, C, 1, 0)$; $(i, P, S_{gq}, C, 0, 1)$; $(i, P, S_{gq}, C, 1, 0)$; $(i, P, P, T, 0, 0)$; $(i, S, S, I, 0, 0)$; $(i, S, P, C, 0, 1)$; $(i, S, P, C, 1, 0)$; $(i, S, P, C, 1, 1)$; $(i, S, S_{gq}, C, 0, 1)$; $(i, S, S_{gq}, C, 1, 0)$; $(i, S, S_{gq}, C, 0, 1)$; $(i, S, S_{gq}, C, 1, 0)$; $(i, S, S, T, 0, 0)$; and thus $M = 18$. Following this pseudo-lexicographic ordering, the infinitesimal generator matrix $\mathbf{Q}$ of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ has the

block tridiagonal structure

$$\mathbf{Q} = \begin{bmatrix} \mathbf{B}_{00} & \mathbf{B}_{01} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{B}_{10} & \mathbf{B}_{11} & \mathbf{A}_1 & \mathbf{0} & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{A}_{-1} & \mathbf{A}_0 & \mathbf{A}_1 & \mathbf{0} & \cdots \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{-1} & \mathbf{A}_0 & \mathbf{A}_1 & \ddots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \ddots \end{bmatrix}, \quad (15)$$

where $\mathbf{B}_{00}$, $\mathbf{B}_{01}$, $\mathbf{B}_{10}$, $\mathbf{B}_{11}$ and $\mathbf{A}_i$, $i \in \{-1, 0, 1\}$, are finite matrices of dimensions 1×1 , $1 \times M$, $M \times 1$, $M \times M$ and $M \times M$, respectively. By definition, since $\mathbf{Q}$ is a generator matrix, we have that $\mathbf{B}_{01}$, $\mathbf{B}_{10}$, $\mathbf{A}_{-1}$ and $\mathbf{A}_1$ are nonnegative matrices while $\mathbf{B}_{00}$, $\mathbf{B}_{11}$ and $\mathbf{A}_0$ have nonnegative offdiagonal elements and strictly negative diagonal elements such that the row sums of $\mathbf{Q}$ equal 0. Moreover, we have that $\mathbf{B}_0 = \mathbf{B}_{00} + \mathbf{B}_{01}$, $\mathbf{B}_1 = \mathbf{B}_{10} + \mathbf{B}_{11} + \mathbf{A}_1$, and $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ are generator matrices.

The matrices corresponding to the non-homogeneous boundary of the state space $\hat{\Omega}$ are given in Appendix C together with additional details. Similarly, the matrices for the homogeneous portion of the state space are given in Appendix C together with additional details.

Following the pseudo-lexicographic ordering of the elements of each level i of the state space, define

$$\boldsymbol{\pi} := (\boldsymbol{\pi}_0, \boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \dots), \quad \boldsymbol{\pi}_0 := (\hat{\pi}(x_0)), \quad \hat{\pi}(x_0) := \lim_{t \rightarrow \infty} \mathbb{P}[X(t) = x_0], \quad (16)$$

$$\boldsymbol{\pi}_i := (\hat{\pi}(x_{i,1}), \dots, \hat{\pi}(x_{i,M})), \quad \hat{\pi}(x_{i,m}) := \lim_{t \rightarrow \infty} \mathbb{P}[X(t) = x_{i,m}], \quad i \in \mathbb{Z}_{\geq 1}, \quad m \in [M]. \quad (17)$$

Assuming the Markov process $\{X(t); t \in \mathbb{R}_+\}$ to be irreducible and ergodic, the limiting probability vector $\boldsymbol{\pi}$ is the stationary distribution of this stochastic process such that its invariant probability vector $\boldsymbol{\pi}$ exists and is uniquely determined as the solution of $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}$ and $\boldsymbol{\pi}\mathbf{e} = 1$, where $\mathbf{e} = (1, \dots, 1)^\top$ [12, 36]. Let $\text{sp}(\mathbf{Y})$ denote the spectral radius of a matrix $\mathbf{Y}$, and let a solution $\mathbf{Y}$ be called a minimal nonnegative solution of a matrix equation if $\mathbf{0} \leq \mathbf{Y} \leq \mathbf{Z}$ for any other nonnegative solution $\mathbf{Z}$, with the inequalities holding componentwise. The stationary probability vector $\boldsymbol{\pi}$ of the process $\{X(t); t \in \mathbb{R}_+\}$ with generator $\mathbf{Q}$ then can be obtained from the following matrix-analytic results.

Theorem 3 (Theorems 1.7.1 and 1.5.1 [48]). *Suppose that the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible and aperiodic with $\mathbf{Q}$ in the form of (15). Further suppose the generator matrix $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ is irreducible. Then this stochastic process is positive recurrent if and only if $\text{sp}(\mathbf{R}) < 1$, where $\mathbf{R}$ is the minimal nonnegative solution of the quadratic matrix equation $\mathbf{R}^2\mathbf{A}_{-1} + \mathbf{R}\mathbf{A}_0 + \mathbf{A}_1 = \mathbf{0}$. Furthermore, there exists a positive vector $(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ such that $(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1) \begin{bmatrix} \mathbf{B}_{00} & \mathbf{B}_{01} \\ \mathbf{B}_{10} & \mathbf{B}_{11} + \mathbf{R}\mathbf{A}_{-1} \end{bmatrix} = \mathbf{0}$. The remaining elements of the invariant probability vector $\boldsymbol{\pi}$ are then given by $\boldsymbol{\pi}_{n+1} = \boldsymbol{\pi}_1\mathbf{R}^n$, $n \in \mathbb{Z}_+$, where $(\boldsymbol{\pi}_0, \boldsymbol{\pi}_1)$ is normalized by $\boldsymbol{\pi}_0\mathbf{e} + \boldsymbol{\pi}_1(\mathbf{I} - \mathbf{R})^{-1}\mathbf{e} = 1$.*

We have above assumed the irreducibility and ergodicity of the Markov process $\{X(t); t \in \mathbb{R}_+\}$. It readily follows from the definition of $\{X(t); t \in \mathbb{R}_+\}$ and its generator matrix $\mathbf{Q}$ that the Markov process is irreducible and aperiodic. Then the stability conditions for the Markov process to be positive recurrent are provided in Theorem 3, under the additional assumption that the generator matrix $\mathbf{A} = \mathbf{A}_{-1} + \mathbf{A}_0 + \mathbf{A}_1$ is irreducible. The irreducibility of $\mathbf{A}$ can be readily verified, as the directed graph corresponding to $\mathbf{A}$ is strongly connected with transitions from state $(*, P, P, I, 0, 0)$ to state $(*, P, P, T, 0, 0)$ reaching every state $(*, P, *, C, *, *)$ in between, transitions from state $(*, S, S, I, 0, 0)$ to state $(*, S, S, T, 0, 0)$ reaching every state $(*, S, *, C, *, *)$ in between, and transitions from states $(*, *, *, T, 0, 0)$ to states $(*, *, *, I, 0, 0)$ connecting the states across both application classes. Since the generator matrix $\mathbf{A}$ of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible, we then have necessary and sufficient conditions for $\text{sp}(\mathbf{R}) < 1$ from the following matrix-analytic result. The stochastic process is therefore positive recurrent if and only if this condition is satisfied.

Theorem 4 (Theorems 1.7.1 [48]). *Suppose that the Markov process $\{X(t); t \in \mathbb{R}_+\}$ is irreducible. Further suppose that the generator $\mathbf{A}$ is irreducible. Then $\text{sp}(\mathbf{R}) < 1$ if and only if $\mathbf{y}\mathbf{A}_1\mathbf{e} < \mathbf{y}\mathbf{A}_{-1}\mathbf{e}$, where $\mathbf{y}$ is the stationary probability vector of $\mathbf{A}$, i.e., the (unique) solution of $\mathbf{y}\mathbf{A} = \mathbf{0}$ and $\mathbf{y}\mathbf{e} = 1$.*

Once the stationary distribution is determined, various functions of the elements of the invariant probability vector $\boldsymbol{\pi}$ can be used to obtain performance measures and quantities of interest. In particular, the tail distribution of the overall queue length process for the QCSC system can be expressed as $\mathbb{P}[Q > z] = \sum_{n=z+1}^{\infty} \boldsymbol{\pi}_n \mathbf{e} = \boldsymbol{\pi}_1 \mathbf{R}^z (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e}$, $z \in \mathbb{Z}_+$, with the corresponding expectation given by $\mathbb{E}[Q] = \sum_{n=0}^{\infty} (n+1) \boldsymbol{\pi}_{n+1} \mathbf{e} = \boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e} + \boldsymbol{\pi}_1 \mathbf{R} (\mathbf{I} - \mathbf{R})^{-2} \mathbf{e}$. The expected sojourn time of a job can then be calculated using Little's law [73] and the above equation, which yields $\mathbb{E}[T] = \lambda^{-1} (\boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \mathbf{e} + \boldsymbol{\pi}_1 \mathbf{R} (\mathbf{I} - \mathbf{R})^{-2} \mathbf{e})$. Additional performance measures and quantities of interest are presented in Appendix C.

5 Optimization of Resource Allocation

As an alternative to the reservation-based resource allocation policy of the previous section, we consider in this section a stochastic network that explicitly represents the QCSC application and system models of Section 2. A primary goal of this paper is to determine the optimal allocation of resource capacity within the context of QCSC environments. The fundamental tradeoff here is between the performance experienced by jobs submitted to the QCSC environment and the financial measures encountered by the host of the environment. We shall focus on the expected sojourn times at the queueing systems of the stochastic network as the performance metric of interest and shall focus on the per-server costs of the resource capacities as the financial metric of interest, noting that related performance and financial metrics can be addressed in a similar manner. Since some queueing systems of the stochastic network may be more important than others, we also allow a weight to be assigned to each queueing system. Then, our formulation of the resource capacity allocation optimization problem of interest consists of minimizing the sum of the weighted expected

sojourn times at each queueing system in the stochastic network subject to a budgetary cost constraint on the resource capacities and subject to the stability constraint of the stochastic network. Alternatively, we could similarly consider an optimization formulation based on minimizing the cost of the resource capacities subject to a constraint on the sum of the weighted expected sojourn times.

More formally, let N denote the number of single-server queues and $\hat{N}$ the number of queueing systems that comprise the stochastic network. For any ordering of the single-server queues of interest, define the vectors $\boldsymbol{\gamma} := (\gamma_1, \dots, \gamma_N)$, $\hat{\boldsymbol{\mu}} := (\hat{\mu}_1, \dots, \hat{\mu}_N)$, and $\boldsymbol{\mathcal{S}} := (\mathcal{S}_1, \dots, \mathcal{S}_N)$, where γ_i is the effective arrival rate to server i , $\hat{\mu}_i$ is the base service rate of jobs at server i , and $\mathcal{S}_i$ is the resource capacity (speed) of the processing unit at server $i \in [N]$. Let $\mathcal{C}$ denote the total budgetary cost constraint on the resource capacities, and $\mathbb{E}[T_i]$ the expected sojourn time at queueing system $i \in [\hat{N}]$. Further define the vectors $\mathbf{c} := (c_1, \dots, c_N)$ and $\boldsymbol{\omega} := (\omega_1, \dots, \omega_{\hat{N}})$, where c_i is the cost associated with each unit of resource capacity (speed) $\mathcal{S}_i$ at server $i \in [N]$ and ω_i is the weight associated with the expected sojourn time at queueing system $i \in [\hat{N}]$. The difference between $\hat{N}$ and N accounts for the fact that each fork-join queueing network consists of two single-server queues, where the expected sojourn times $\mathbb{E}[T_i]$ and sojourn-time weights ω_i are associated with the queueing systems $i \in [\hat{N}]$; and where the effective arrival rates γ_i , base service rates $\hat{\mu}_i$, resource capacities $\mathcal{S}_i$, and per-unit resource-capacity costs c_i are associated with the single-server queues $i \in [N]$.

The objective function consists of minimizing the sum of weighted expected sojourn times, and is thus defined over the $\hat{N}$ queueing systems composing the stochastic network. Meanwhile, the budgetary cost constraint involves the sum of cost-weighted resource capacities, and thus is defined together with the stability constraints over the N single-server queues composing the stochastic network. We therefore formulate our resource capacity allocation optimization problem based on the queueing systems of the stochastic network, while at the same time ensuring that the budgetary cost constraint and the stability constraints include both servers for each fork-join queueing network (following the approach described below). More specifically, our resource capacity allocation optimization problem is formulated as:

$$(\text{OPT}_{\text{RCA}}) \quad \min_{\boldsymbol{\mathcal{S}} \in (0, \infty)^{\hat{N}}} \sum_{i=1}^{\hat{N}} \omega_i \mathbb{E}[T_i] \quad \text{s.t.} \quad \sum_{i=1}^{\hat{N}} c_i \mathcal{S}_i \leq \mathcal{C}, \quad \text{and} \quad \mathcal{S}_i \hat{\mu}_i > \gamma_i, \quad \forall i \in [\hat{N}]. \quad (18)$$

The difficulty in solving $(\text{OPT}_{\text{RCA}})$ is rooted in determining the expected sojourn times $\mathbb{E}[T_i]$ as a function of the resource capacities $\boldsymbol{\mathcal{S}}$ and base service rates $\boldsymbol{\mu}$. As described in Section 3.1, we know that the sequential computational phase represents a separable product-form queueing network. In addition, a sequential computational phase followed by the fork-join queueing network of a parallel computational phase would also represent a separable queueing network, given the Poisson departure process from the sequential computational phase due to Burke's Theorem [13] and quasireversibility [28, 44]; see Section 3.1. However, even though the service times at each queue follow an exponential distribution,

the application workload model comprising combinations of both the parallel-sequential application class and the sequential-parallel application class does not ensure Poisson arrival processes to each queue of the stochastic network. The main difficulty is not knowing the higher-order properties of the arrival process to each queueing system i beyond its effective arrival rate γ_i , $i \in [\hat{N}]$; moreover, these properties of the arrival process depend on the resource capacities of the single-server queues $j \in [N]$ that feed jobs to queueing system $i \in [\hat{N}]$ in the stochastic network. Hence, a standard approach for solving optimization problems such as $(\text{OPT}_{\text{RCA}})$ is based on a combination of simulation methods and gradient methods, such as stochastic approximation [4, 33, 45]. Although an effective approach to solve problems such as $(\text{OPT}_{\text{RCA}})$, stochastic approximation can be computationally expensive in both the time and resources required to obtain optimal solutions in practice for problems involving large-scale stochastic networks. This is in large part due to the inherent combinatorial explosion of the decision space and the inherent costs of simulation to estimate gradients in stochastic network models of real-world QCSC environments.

To address the computational complexity of stochastic approximation, we follow the approach of Dieker, Ghosh and Squillante [15] within the context of QCSC environments. Observe from (1) and (3), and even (36) and (37) in the proof of Theorem 2, that the expression for the expected sojourn time of queueing system i under any resource capacity allocation $\mathbf{S}$ has a functional form along the lines of $\mathbb{E}[T_i] = \zeta_i / (\mathcal{S}_i \hat{\mu}_i - \gamma_i)$, $i \in [\hat{N}]$. Upon substituting this functional form into the objective function (18) of $(\text{OPT}_{\text{RCA}})$, we obtain the revised optimization problem formulation:

$$(\text{OPT}_{\text{RCA:FF}}) \quad \min_{\mathbf{S} \in (0, \infty)^{\hat{N}}} \sum_{i=1}^{\hat{N}} \omega_i \frac{\zeta_i}{\mathcal{S}_i \hat{\mu}_i - \gamma_i} \quad \text{s.t.} \quad \sum_{i=1}^{\hat{N}} c_i \mathcal{S}_i \leq \mathcal{C}, \quad \text{and} \quad \mathcal{S}_i \hat{\mu}_i > \gamma_i, \quad \forall i \in [\hat{N}] \quad (19)$$

Then, given the vector $\boldsymbol{\zeta} := (\zeta_1, \dots, \zeta_{\hat{N}})$, the optimization problem $(\text{OPT}_{\text{RCA:FF}})$ can be directly solved as established in the following theoretical result.

Theorem 5. *The solution of the resource capacity allocation optimization problem formulation $(\text{OPT}_{\text{RCA:FF}})$ is given by the optimal resource capacity vector $\mathbf{S}^* := (\mathcal{S}_1^*, \dots, \mathcal{S}_{\hat{N}}^*)$ where*

$$\mathcal{S}_i^* = \frac{\gamma_i}{\hat{\mu}_i} + \left(\mathcal{C} - \sum_{j=1}^{\hat{N}} c_j \frac{\gamma_j}{\hat{\mu}_j} \right) \frac{\sqrt{\omega_i \zeta_i / (c_i \hat{\mu}_i)}}{\sum_{j=1}^{\hat{N}} \sqrt{\omega_j \zeta_j c_j / \hat{\mu}_j}}. \quad (20)$$

An analogous result was obtained by Kleinrock in the context of service rates of multiple channel systems [31] and by Wein in the context of service rates of the Brownian approximation of generalized Jackson networks [70].

Based on the result of Theorem 5, we consider the following fixed-point iterative process for the solution of $(\text{OPT}_{\text{RCA:FF}})$ where the difference between $\hat{N}$ and N is supported by our special functional form for the expected sojourn time of each fork-join queueing network,

which further guarantees the budgetary cost constraint and the stability constraints properly include both servers of the fork-join queueing system. This special functional form is defined after we present our fixed-point iterative process within which it plays an important role.

Step 0: Initialize the unknown variables $\zeta_i^{(0)}$ for each queueing system $i \in [\hat{N}]$, and set $k = 0$.

Ming-Hung: For *Step 0*, you should have one $\zeta_i^{(0)}$ for each non-FJ queue i ; and you should have one $\zeta_j^{(0)}$ for each FJ queue j .

Step 1: Then, from (20), compute the optimal resource capacity vector $\mathbf{S}^{(k+1)}$ for fixed-point iteration $k + 1 \in \mathbb{N}$, conditional on the vector $\boldsymbol{\zeta}^{(k)}$, with iterates in terms of γ_i , $\hat{\mu}_i$, c_i and ω_i for all $i \in [\hat{N}]$ given by

$$\mathcal{S}_i^{(k+1)} = \frac{\gamma_i}{\hat{\mu}_i} + \left(c - \sum_{j=1}^{\hat{N}} c_j \frac{\gamma_j}{\hat{\mu}_j} \right) \frac{\sqrt{\omega_i \zeta_i^{(k)} / (c_i \hat{\mu}_i)}}{\sum_{j=1}^{\hat{N}} \sqrt{\omega_j \zeta_j^{(k)}} c_j / \hat{\mu}_j}, \quad i \in [\hat{N}], \quad k + 1 \in \mathbb{N}. \quad (21)$$

We define below the handling of the base service rate $\hat{\mu}_i$ and the cost c_i as part of the special functional form for each fork-join queueing system $i \in [\hat{N}]$ with respect to the base service rates and costs of its two servers.

Ming-Hung: For *Step 1*, you should have γ_i , $\hat{\mu}_i$, c_i , $\zeta_i^{(k)}$ for each non-FJ queue i ; and you should have γ_j , $\hat{\mu}_j$ set equal to $\hat{\mu}_1$ for the bottleneck server, c_j set equal to $c_{FJ} = c_1 + c_2/r$, $\zeta_j^{(k)}$ for each FJ queue j .

Step 2: Simulate the stochastic network under the resource capacity vector $\mathbf{S}^{(k+1)}$ to obtain the corresponding expected sojourn time $\mathbb{E}[T_i^{(k+1)}]$ for each queueing system $i \in [\hat{N}]$ in terms of $\hat{\mu}_j$ for all $j \in [N]$. The handling of the resource capacities and base service rates for both servers of each fork-join queueing system $i \in [\hat{N}]$ is defined below as part of its special functional form with respect to the resource capacity $\mathbf{S}_i^{k+1}$ from *Step 1*.

Ming-Hung: For *Step 2*, you should have $\mathcal{S}_i$ set equal to the corresponding result $\mathcal{S}_i^{(k+1)}$ from (21), γ_i , $\hat{\mu}_i$ for each non-FJ queue i ; and you should have $\mathcal{S}_1$ for the bottleneck server set equal to the corresponding result $\mathcal{S}_j^{(k+1)}$ from (21), $\mathcal{S}_2$ for the non-bottleneck server set equal to the corresponding result $\mathcal{S}_j^{(k+1)}/r$ from (21), γ_j , $\hat{\mu}_1$ for the bottleneck server, $\hat{\mu}_2$ for the non-bottleneck server, for each FJ queue j .

Step 3: Given the expected sojourn times $\mathbb{E}[T_i^{(k+1)}]$ from *Step 2*, the functional form for each queueing system $i \in [\hat{N}]$ is inverted to yield

$$\zeta_i^{(k+1)} = \mathbb{E}[T_i^{(k+1)}](\mathcal{S}_i^{(k+1)} \hat{\mu}_i - \gamma_i), \quad i \in [\hat{N}], \quad k + 1 \in \mathbb{N}, \quad (22)$$

where the handling of the resource capacity $\mathcal{S}_i^{k+1}$ and base service rate $\hat{\mu}_i$ of each fork-join queueing system $i \in [\hat{N}]$ is defined below as part of its special functional form.

Ming-Hung: For *Step 3*, you should have $\mathbb{E}[T_i^{(k+1)}]$ set equal to the corresponding result from simulation, γ_i , $\hat{\mu}_i$, $\mathcal{S}_i^{(k+1)}$ from (21), for each non-FJ queue i ; and you should have $\mathbb{E}[T_j^{(k+1)}]$ set equal to the corresponding result from simulation, γ_j , $\hat{\mu}_j$ set equal to $\hat{\mu}_1$ for the bottleneck server, $\mathcal{S}_i^{(k+1)}$ set equal to $\mathcal{S}_1$ for the bottleneck server, for each FJ queue j .

Step 4: Then, test for the convergence of the fixed-point solution up to a desired level of

accuracy, say, ensuring $\mathcal{S}_i^{(k+1)} - \mathcal{S}_i^{(k)} < \epsilon$ for all $i \in [\hat{N}]$. If not, increment k by one and repeat the fixed-point iterative process from *Step 1*.

Step 5: Output the optimal resource capacity vector $\mathcal{S}^*$ and the corresponding expected sojourn times $\mathbb{E}[T_i]$, $i \in [\hat{N}]$.

Dieker, Ghosh and Squillante [15] analyze certain theoretical properties of a fixed-point iterative process similar to *Steps 0 – 5* above within the context of Brownian tree networks and without the inclusion of the more complex fork-join queueing systems. We now turn to consider the theoretical properties of our foregoing fixed-point iterative process based on (21) and (22) within the context of a Brownian approximation of a generalized Jackson network [14] — with general interarrival time distributions and exponential service time distributions — for our stochastic network comprising the QCSC application and system models of Section 2. The latter Brownian approximation of a generalized Jackson network for our stochastic network includes both a non-tree network topology and combinations of tandem and fork-join queueing networks in parallel and in series.

Functional Form for Fork-Join Queueing Networks. Let us first consider the special functional form for the expected sojourn time of each fork-join queueing network which plays an important role in the context of our fixed-point iterative process as noted above. Recall that $\hat{\mu}_1 \leq \hat{\mu}_2$ and $r := \hat{\mu}_2/\hat{\mu}_1 \in [1, \infty)$ for the fork-join queueing network of interest. Let $\mathcal{S}_1$ and $\mathcal{S}_2$ respectively denote the resource capacity (speed) of the processing unit associated with $\hat{\mu}_1$ and $\hat{\mu}_2$. Based on common aspects of our sojourn-time approximations in Section 3.2, especially our light-heavy traffic interpolation approximations (13) and (14) as well as the exact solution (3) when $r = 1$, the functional form for the expected sojourn time of the fork-join queueing network under the resource capacity allocation $\mathcal{S}_1$ for the bottleneck server can be expressed as $\mathbb{E}[T_{FJ}] = \zeta_{FJ}/(\mathcal{S}_1\hat{\mu}_1 - \gamma_{FJ})$. Hence, the fixed-point iterative process directly yields the optimal resource capacity for $\mathcal{S}_1$. At the same time, we need to determine the resource capacity allocation $\mathcal{S}_2$ for the non-bottleneck server from the fixed-point iterative solution for $\mathcal{S}_1$ in a manner that takes into account both r and the combined cost of each server. To this end, let c_1 and c_2 respectively denote the cost associated with each unit of resource capacity $\mathcal{S}_1$ and $\mathcal{S}_2$. The combined cost for the fork-join queueing network clearly must include c_1 for the resource capacity $\mathcal{S}_1$. Since the impact of the non-bottleneck server quickly declines as $\hat{\mu}_2$ deviates from $\hat{\mu}_1$, the combined cost should also include a portion of c_2 that decreases as $\hat{\mu}_2/\hat{\mu}_1$ increases. We therefore define the cost of the fork-join queueing network as part of the fixed-point iterative process to be $c_{FJ} := c_1 + (\hat{\mu}_1/\hat{\mu}_2)c_2 = c_1 + c_2/r$.

Each iteration of the fixed-point iterative process then proceeds as follows. From *Step 1*, the optimal resource capacity $\mathcal{S}_{FJ}^*$ for the fork-join queueing network is given by (21) in terms of c_{FJ} , ω_{FJ} and the functional form for $\mathbb{E}[T_{FJ}]$ above, together with the variables for the rest of the stochastic network. Now we can equate the total cost of the fork-join queueing network, i.e., $\mathcal{S}_1c_1 + \mathcal{S}_2c_2$, with the optimal cost from the fixed-point iterative process, i.e., $\mathcal{S}_{FJ}^*c_{FJ}$, in order to obtain $\mathcal{S}_2$, which further maintains a similar logical argument for $\mathcal{S}_2$ as above for c_2 in c_{FJ} . In particular, upon setting $\mathcal{S}_1 = \mathcal{S}_{FJ}^*$, the corresponding resource

capacity allocation $\mathcal{S}_2$ can be derived as

$$\begin{aligned}\mathcal{S}_1 c_1 + \mathcal{S}_2 c_2 &= \mathcal{S}_{FJ}^* c_{FJ} = \mathcal{S}_1 (c_1 + c_2/r) = \mathcal{S}_1 c_1 + \mathcal{S}_1 (c_2/r), \\ \mathcal{S}_2 &= \mathcal{S}_1/r,\end{aligned}$$

which leads to a resource capacity allocation for the fork-join queueing network that maintains $\mathcal{S}_2 \hat{\mu}_2 = \mathcal{S}_1 \hat{\mu}_1$. Next, $\mathcal{S}_1$ and $\mathcal{S}_2$ together with $\hat{\mu}_1$, $\hat{\mu}_2$ and γ_{FJ} are used for the fork-join queueing network in the simulation of the entire stochastic network as part of *Step 2*. The inverted functional form for the expected sojourn time of the fork-join queueing network in *Step 3* yields $\zeta_{FJ} = \mathbb{E}[T_{FJ}](\mathcal{S}_1 \hat{\mu}_1 - \gamma_{FJ})$. Then, the fixed-point iterative process continues until the convergence test in *Step 4* is satisfied, at which point the iterative process concludes with *Step 5*. We noted above that our special functional form for fork-join queueing networks ensures that the budgetary cost constraint and the stability constraints are properly captured in the formulation (OPT_{RCA:FF}). To see this, observe that $c_{FJ} \mathcal{S}_1 = \mathcal{S}_1 (c_1 + c_2/r) = \mathcal{S}_1 c_1 + \mathcal{S}_1 c_2/r = \mathcal{S}_1 c_1 + \mathcal{S}_2 c_2$; moreover, $\mathcal{S}_1 \hat{\mu}_1 > \gamma_{FJ}$ guarantees that $\mathcal{S}_2 \hat{\mu}_2 > \gamma_{FJ}$ since server 1 by definition is the bottleneck server. Hence, we have $\sum_{i=1}^N c_i \mathcal{S}_i = \sum_{i=1}^N c_i \mathcal{S}_i \leq \mathcal{C}$ as well as $\mathcal{S}_i \hat{\mu}_i > \gamma_i$ for all $i \in [N]$.

Our above approach for the functional form of the expected sojourn time of each fork-join queueing network within the context of our fixed-point iterative process is also consistent with our theoretical results and observations in Section 3.2 and subsequently in Section 6.2. More precisely, in the limit as $r \rightarrow \infty$, we have that c_{FJ} asymptotically converges to c_1 and $\mathcal{S}_2$ asymptotically converges to 0, thus rendering $\mathcal{S}_2 \hat{\mu}_2 \rightarrow 0$. These asymptotic limits are appropriately consistent with the situations where the bottleneck processing unit associated with $\hat{\mu}_1$ increasingly dominates the expected sojourn time of the fork-join queueing network while the impact of the non-bottleneck processing unit associated with $\hat{\mu}_2$ rapidly declines, both as r increases. On the other hand, in the limit as $r \rightarrow 1$, we have that c_{FJ} asymptotically converges to $c_1 + c_2$ and $\mathcal{S}_2$ asymptotically converges to $\mathcal{S}_1$, thus rendering $\mathcal{S}_2 \hat{\mu}_2 \rightarrow \mathcal{S}_1 \hat{\mu}_1$. These asymptotic limits are appropriately consistent with the situations where the non-bottleneck processing unit associated with $\hat{\mu}_2$ increasingly impacts the expected sojourn time of the fork-join queueing network together with the bottleneck processing unit associated with $\hat{\mu}_1$ as r increases up to the point at which, in the absence of a bottleneck server, both processing units play a relatively equal role in the expected sojourn time of the fork-join queueing network. Finally, it is important to note that these results on c_{FJ} and on $\mathcal{S}_1$ and $\mathcal{S}_2$ from our above approach have nice physical interpretations in themselves. More specifically, since r relates the inherent service rates $\hat{\mu}_1$ and $\hat{\mu}_2$ of the bottleneck and non-bottleneck servers with a single base unit, this heterogeneity ratio should similarly relate the overall service rates $\mathcal{S}_1 \hat{\mu}_1$ and $\mathcal{S}_2 \hat{\mu}_2$ under the resource capacity allocations as the number of resource capacity units $\mathcal{S}_1$ and $\mathcal{S}_2$ change for the bottleneck or non-bottleneck servers. Such a physical interpretation analogously holds for the cost c_{FJ} of the fork-join queueing network as part of the fixed-point iterative process.

Brownian Approximation of Generalized Jackson Networks Next let us consider a Brownian approximation of a generalized Jackson network with general interarrival time

distributions and exponential service time distributions for our stochastic network. This serves as the basis of our mathematical analysis to establish the key theoretical properties of our fixed-point iterative process that takes into account the higher-order properties of the arrival process to each queue i beyond its effective arrival γ_i , $i \in [\hat{N}]$.

The topology of the stochastic network for our QCSC application and system models of Section 2 can be characterized by a graph $G = (V, E)$, where the set of vertices $V = [\hat{N}]$ represents the queues of the stochastic network, and the set of (directed) edges E represents the paths of jobs from one queue to another queue. The $\hat{N}$ vertices of V are ordered such that, for each directed edge $(i, j) \in E$, i is always strictly less than j , $i, j \in [\hat{N}]$. Since arriving jobs belong to the parallel-sequential application class with probability $p_{\parallel}$ and belong to the sequential-parallel application class with probability $1 - p_{\parallel}$, we define the root for the parallel-sequential application class to be queue 1, represented by vertex 1 in G , and similarly define the root for the sequential-parallel application class to be queue 2, represented by vertex 2 in G . For any queue $i \in [\hat{N}]^- := [\hat{N}] \setminus \{1, 2\}$, we denote its root vertex in the graph G by $\varpi(i)$.

Let $p_{i,j}$ denote the probability associated with the edge from vertex i to vertex j , $i < j$, and thus $p_{i,j} = 0$ if there is no edge between i and j . Further let $\iota(i)$ denote the parent vertex of vertex i , and $\mathcal{P}(i) = (\varpi(i), \dots, \iota(i), i)$ the (unique) path taken by a job from its root to vertex i in the stochastic network, $i \in [\hat{N}]^-$. For any $j \in \mathcal{P}(i)$, define $\mathcal{Q}(i, j) := \prod_{\ell \in \mathcal{P}(i), \ell \neq j} p_{\iota(\ell), \ell}$ with $\prod_{\emptyset} := 1$; and for any vector $\mathbf{z}$, define $\mathbf{z}_i$ to be the vector with $\mathbf{z}$ confined to those elements in the path $\mathcal{P}(i)$, $i \in [\hat{N}]$. Given the feedforward structure of the stochastic network topology, the expected sojourn time at queue i depends only on the vector $\hat{\mathbf{S}}_i = (\mathcal{S}_{\varpi(i)} \hat{\mu}_{\varpi(i)}, \dots, \mathcal{S}_{\iota(i)} \hat{\mu}_{\iota(i)}, \mathcal{S}_i \hat{\mu}_i)$ and thus we denote this expected sojourn time by $\mathbb{E}[T_i(\hat{\mathbf{S}}_i)]$ with $\mathbb{E}[T_j(\hat{\mathbf{S}}_i)]$ for $j \in \mathcal{P}(i)$, $i \in [\hat{N}]$. Further define $\kappa_i(\mathbf{z}_i) := \sum_{j \in \mathcal{P}(i)} \mathcal{Q}(i, j) \mathbb{E}[T_j(\gamma_i + \mathbf{z}_i)]$, from which we obtain

$$\mathbb{E}[T_i(\hat{\mathbf{S}}_i)] = \kappa_i(\hat{\mathbf{S}}_i - \gamma_i) - p_{\iota(i), i} \kappa_{\iota(i)}(\hat{\mathbf{S}}_{\iota(i)} - \gamma_{\iota(i)}). \quad (23)$$

Then, upon substituting (23) into the objective function (19) of $(\text{OPT}_{\text{RCA:FF}})$, we obtain the following revised optimization problem formulation in terms of the functions $\kappa_i(\cdot)$:

$$(\text{OPT}_{\text{RCA:FF:}\kappa}) \quad \min_{\mathbf{S} \in (0, \infty)^{\hat{N}}} \sum_{i=1}^{\hat{N}} \omega_i (\kappa_i(\hat{\mathbf{S}}_i - \gamma_i) - p_{\iota(i), i} \kappa_{\iota(i)}(\hat{\mathbf{S}}_{\iota(i)} - \gamma_{\iota(i)})) \quad \text{s.t.} \quad \sum_{i=1}^{\hat{N}} c_i \mathcal{S}_i \leq \mathcal{C}, \quad \mathcal{S}_i \hat{\mu}_i > \gamma_i, \quad \forall i \in [\hat{N}] \quad (24)$$

The optimization problem $(\text{OPT}_{\text{RCA:FF:}\kappa})$ yields several favorable structural properties that we establish in the following theoretical results.

Lemma 5. *For any $i \in [\hat{N}]$, the function $\kappa_i : (0, \infty)^{|\mathcal{P}(i)|} \rightarrow \mathbb{R}_+$ has the following properties: (1) it is convex on its domain; (2) it is nonincreasing in each coordinate; (3) it is strictly decreasing in the last coordinate; (4) it satisfies $\delta \kappa_i(\delta \mathbf{z}_i) = \kappa_i(\mathbf{z}_i)$, for $\mathbf{z}_i > 0$ and $\delta > 0$.*

The first (convexity) property yields a convex program when the weights are nonincreasing, i.e., $\omega_1 \geq \dots \geq \omega_{\hat{N}}$, and otherwise renders a (weighted) difference of convex functions

programming problem [2]. The second (monotonicity) property implies that the expected sojourn time $\mathbb{E}[T_i(\hat{\mathbf{S}}_i)]$ at queue i decreases in the resource capacity service rate $\mathcal{S}_i \hat{\mu}_i$. The last property implies that $\kappa_i(\cdot)$ is homogeneous, which follows from the Brownian scaling property.

Existence and Uniqueness of Fixed-Point Let us now consider the theoretical properties of our fixed-point iterative process within the context of a Brownian approximation of a generalized Jackson network for our stochastic network. We start by establishing the existence and uniqueness of the fixed-point for our iterative process.

Define $\zeta_i(\hat{\mathbf{S}}_i) := (\mathcal{S}_i \hat{\mu}_i - \gamma_i) \mathbb{E}[T_i(\hat{\mathbf{S}}_i)]$, $i \in [\hat{N}]$, and thus the system of nonlinear fixed-point equations resulting from Theorem 5 is given by

$$\mathcal{S}_i^* = \frac{\gamma_i}{\hat{\mu}_i} + \left(\mathcal{C} - \sum_{j=1}^N c_j \frac{\gamma_j}{\hat{\mu}_j} \right) \frac{\sqrt{\omega_i \zeta_i(\hat{\mathbf{S}}^*) / (c_i \hat{\mu}_i)}}{\sum_{j=1}^N \sqrt{\omega_j \zeta_j(\hat{\mathbf{S}}^*) c_j / \hat{\mu}_j}}, \quad i \in [\hat{N}]. \quad (25)$$

Further define $\hat{\mathbf{z}}(\hat{\mathbf{S}}) := (\hat{z}_1(\hat{\mathbf{S}}), \hat{z}_2(\hat{\mathbf{S}}), \dots, \hat{z}_{\hat{N}}(\hat{\mathbf{S}}))$ and $\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i) := (\hat{z}_{\varpi(i)}(\hat{\mathbf{S}}_{\varpi(i)}), \dots, \hat{z}_{\iota(i)}(\hat{\mathbf{S}}_{\iota(i)}), \hat{z}_i(\hat{\mathbf{S}}_i))$ where $\hat{\mathbf{S}} = (\mathcal{S}_1 \hat{\mu}_1, \mathcal{S}_2 \hat{\mu}_2, \dots, \mathcal{S}_{\hat{N}} \hat{\mu}_{\hat{N}})$, $\hat{z}_i(\hat{\mathbf{S}}) := \hat{z}_i(\hat{\mathbf{S}}_i) := (\mathcal{S}_i \hat{\mu}_i - \gamma_i) / (\mathcal{S}_{\varpi(i)} \hat{\mu}_{\varpi(i)} - \gamma_{\varpi(i)})$, $\hat{z}_1(\hat{\mathbf{S}}) := 1$, and $\hat{z}_2(\hat{\mathbf{S}}) := 1$. Additionally define $\mathbb{S} := \{\hat{\mathbf{S}} \in (\gamma_1, \infty) \times \dots \times (\gamma_{\hat{N}}, \infty) : \sum_{i=1}^{\hat{N}} c_i \mathcal{S}_i \leq \mathcal{C}\}$. We then have the following theoretical result on an important one-to-one mapping.

Lemma 6. *The mapping from $\hat{\mathbf{S}} \in \mathbb{S}$ to $\hat{\mathbf{z}}(\hat{\mathbf{S}}) \in \{1\} \times \{1\} \times \mathbb{R}_+^{\hat{N}-2}$ is one-to-one.*

From (23), the definitions of $\zeta_i(\hat{\mathbf{S}}_i)$ and $\hat{z}_i(\hat{\mathbf{S}}_i)$ above, and the homogeneity property of $\kappa(\cdot)$ from Lemma 5, we derive

$$\begin{aligned} \zeta_i(\hat{\mathbf{S}}) &= (\mathcal{S}_i \hat{\mu}_i - \gamma_i) \mathbb{E}[T_i(\hat{\mathbf{S}})] = \hat{z}_i(\hat{\mathbf{S}}_i) (\mathcal{S}_{\varpi(i)} \hat{\mu}_{\varpi(i)} - \gamma_{\varpi(i)}) (\kappa_i(\hat{\mathbf{S}}_i - \gamma_i) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{S}}_{\iota(i)} - \gamma_{\iota(i)})) \\ &= \hat{z}_i(\hat{\mathbf{S}}_i) (\kappa_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i)) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}(\hat{\mathbf{S}}_{\iota(i)}))), \end{aligned} \quad (26)$$

and thus the topology of the stochastic network renders $\zeta_i(\hat{\mathbf{S}})$ to be solely dependent on $\hat{\mathbf{S}}_i$ with the Brownian approximation of the network further reducing the dependency to only $\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i)$. Let us define

$$\zeta_i(\hat{\mathbf{z}}_i) := \hat{z}_i(\kappa_i(\hat{\mathbf{z}}_i) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)})). \quad (27)$$

Upon rewriting the system of fixed-point equations (25) with respect to $\hat{\mathbf{z}}_i$, we have

$$\frac{\mathcal{C} - \sum_{j=1}^{\hat{N}} c_j \frac{\gamma_j}{\hat{\mu}_j}}{c_i (\mathcal{S}_i^* \hat{\mu}_i - \gamma_i)} \sqrt{\omega_i \zeta_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i^*)) c_i \hat{\mu}_i} = \sum_{j=1}^{\hat{N}} \sqrt{\omega_j \zeta_j(\hat{\mathbf{z}}_j(\hat{\mathbf{S}}_j^*)) c_j / \hat{\mu}_j},$$

from which we determine that $\sqrt{\omega_i \zeta_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i^*)) c_i \hat{\mu}_i} / (c_i (\mathcal{S}_i^* \hat{\mu}_i - \gamma_i))$ does not depend on i . It then follows that

$$\frac{\sqrt{\omega_i \zeta_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i^*)) c_i \hat{\mu}_i}}{c_i (\mathcal{S}_i^* \hat{\mu}_i - \gamma_i)} = \frac{\sqrt{\omega_{\varpi(i)} \zeta_{\varpi(i)}(\hat{\mathbf{z}}_{\varpi(i)}(\hat{\mathbf{S}}_{\varpi(i)}^*)) c_{\varpi(i)} \hat{\mu}_{\varpi(i)}}}{c_{\varpi(i)} (\mathcal{S}_{\varpi(i)}^* \hat{\mu}_{\varpi(i)} - \gamma_{\varpi(i)})},$$

or equivalently

$$\frac{\zeta_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i^*))}{\hat{z}_i(\hat{\mathbf{S}}_i^*)} = \zeta_{\varpi(i)} \frac{c_i/\omega_i}{c_{\varpi(i)}/\omega_{\varpi(i)}} \hat{z}_i(\hat{\mathbf{S}}_i^*).$$

Upon rewriting (26) as

$$\kappa_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i^*)) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}(\hat{\mathbf{S}}_{\iota(i)}^*)) = \frac{\zeta_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i^*))}{\hat{z}_i(\hat{\mathbf{S}}_i^*)},$$

we conclude that $\mathbf{S}^*$ satisfies the system of fixed-point equations (25) if and only if

$$\kappa_i(\hat{\mathbf{z}}_i(\hat{\mathbf{S}}_i^*)) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}(\hat{\mathbf{S}}_{\iota(i)}^*)) = \zeta_{\varpi(i)} \frac{c_i/\omega_i}{c_{\varpi(i)}/\omega_{\varpi(i)}} \hat{z}_i(\hat{\mathbf{S}}_i^*), \quad i \in [\hat{N}].$$

We then have the following theoretical result establishing that there is exactly one solution $\mathbf{z}^*$ to the above equation, from which it follows that there is exactly one solution $\mathbf{S}^*$ of the fixed-point equations (25) on the boundary of the feasible set.

Theorem 6. *The system of fixed-point equations (25) has exactly one solution on the set $\mathbb{S}$.*

Convergence of Fixed-Point Iterative Process Lastly, let us consider the theoretical convergence properties of our fixed-point iterative process within the context of a Brownian approximation of a generalized Jackson network for our stochastic network. We begin by analyzing the iterates $\{\hat{\mathbf{S}}^{(k)}\}$ through $\{\hat{\mathbf{z}}^{(k)} = \hat{\mathbf{z}}(\hat{\mathbf{S}}^{(k)})\}$, as defined above to be $\hat{z}_i(\hat{\mathbf{S}}) := \hat{z}_i(\hat{\mathbf{S}}_i) := (\mathcal{S}_i \hat{\mu}_i - \gamma_i) / (\mathcal{S}_{\varpi(i)} \hat{\mu}_{\varpi(i)} - \gamma_{\varpi(i)})$, while exploiting the structural properties of $\kappa_i(\cdot)$.

More specifically, the two sequences $\{\hat{\mathbf{S}}^{(k)}\}$ and $\{\hat{\mathbf{z}}^{(k)} = \hat{\mathbf{z}}(\hat{\mathbf{S}}^{(k)})\}$ are in one-to-one correspondence in light of Lemma 6. The system of fixed-point equations

$$\hat{\mathbf{S}}_i^{(k+1)} = \frac{\gamma_i}{\hat{\mu}_i} + \left(\mathcal{C} - \sum_{j=1}^{\hat{N}} c_j \frac{\gamma_j}{\hat{\mu}_j} \right) \frac{\sqrt{\omega_i \zeta_i(\hat{\mathbf{z}}_i^{(k)}) / (c_i \hat{\mu}_i)}}{\sum_{j=1}^{\hat{N}} \sqrt{\omega_j \zeta_j(\hat{\mathbf{z}}_j^{(k)}) c_j / \hat{\mu}_j}}, \quad i \in [\hat{N}], \quad (28)$$

corresponding to (21) then becomes

$$\hat{z}_i^{(k+1)} = \sqrt{\frac{\omega_i \zeta_i(\hat{\mathbf{z}}_i^{(k)}) / (c_i \hat{\mu}_i)}{\omega_{\varpi(i)} \zeta_{\varpi(i)}(\hat{\mathbf{z}}_{\varpi(i)}^{(k)}) / (c_{\varpi(i)} \hat{\mu}_{\varpi(i)})}}. \quad (29)$$

In particular, the fixed-point iterates in (28) can be equivalently characterized through (29) with $\zeta_i(\hat{\mathbf{z}}_i)$ as defined in (27) where $\hat{z}_1^{(k)} = \hat{z}_2^{(k)} = 1$ for all $k \geq 1$.

Now our analysis of the convergence properties of our fixed-point iterative process establishes that the above iterates move toward the fixed point with the magnitude of each step being larger for iterates further away from the fixed point. To this end, we start by showing the desired theoretical result for iterates corresponding to the children of each root.

Lemma 7. Suppose queue i , $i \in [\hat{N}]^-$, is a child of a root. Then, if $\hat{z}_i^{(k)} < \hat{z}_i^*$, the corresponding iterates satisfy

$$\frac{\hat{z}_i^{(k+1)} - \hat{z}_i^{(k)}}{\hat{z}_i^{(k)}} \geq \sqrt{\frac{\hat{z}_i^*}{\hat{z}_i^{(k)}}} - 1 > 0. \quad (30)$$

Similarly, if $\hat{z}_i^{(k)} > \hat{z}_i^*$, the corresponding iterates satisfy

$$\frac{\hat{z}_i^{(k)} - \hat{z}_i^{(k+1)}}{\hat{z}_i^{(k)}} \geq 1 - \sqrt{\frac{\hat{z}_i^*}{\hat{z}_i^{(k)}}} > 0. \quad (31)$$

Next, we turn to establish this desired theoretical result more generally for any queue in the stochastic network.

Lemma 8. Fixing queue i , $i \in [\hat{N}]^-$, suppose that $\lim_{k \rightarrow \infty} \hat{\mathbf{z}}_{i(i)}^{(k)} = \hat{\mathbf{z}}_{i(i)}^*$. Then, for any $\xi > 0$, there exists a $\delta = \delta(\xi) > 0$ such that (30) holds for $\hat{\mathbf{z}}_i^{(k)} \in \mathcal{L}_i^\xi$ and (31) holds for $\hat{\mathbf{z}}_i^{(k)} \in \mathcal{U}_i^\xi$, where $\mathcal{L}_i^\xi := [\hat{z}_{\varpi(i)}^* \pm \delta(\xi)] \times \cdots \times [\hat{z}_{i(i)}^* \pm \delta(\xi)] \times (0, \hat{z}_i^* - \xi)$ and $\mathcal{U}_i^\xi := [\hat{z}_{\varpi(i)}^* \pm \delta(\xi)] \times \cdots \times [\hat{z}_{i(i)}^* \pm \delta(\xi)] \times (\hat{z}_i^* + \xi, \infty)$.

It is important to note that Lemma 7 and Lemma 8 do not prohibit the potential for oscillations under our fixed-point iterative process, which implies the possibility that the iterates may not converge. To address this possibility and ensure the iterates converge to the fixed point, we modify our fixed-point iterative process ever so slightly as follows. The modified fixed-point iterative process maintains an interval $\mathcal{I}_i^{(k)}$ within which $\hat{z}_i^*$ and all modified iterates $\hat{z}_i^{(k+1)}, \hat{z}_i^{(k+2)}, \dots$ will lie. Here, the left endpoint of $\mathcal{I}_i^{(k)}$ is defined as the largest iterate among $\hat{z}_i^{(0)}, \hat{z}_i^{(1)}, \dots, \hat{z}_i^{(k)}$ that is smaller than $\hat{z}_i^*$; and the right endpoint of $\mathcal{I}_i^{(k)}$ is similarly defined as the smallest iterate among $\hat{z}_i^{(0)}, \hat{z}_i^{(1)}, \dots, \hat{z}_i^{(k)}$ that is larger than $\hat{z}_i^*$. Then, a bisection step is added such that, if $\hat{z}_i^{(k+1)}$ calculated from (29) lies outside of the interval $\mathcal{I}_i^{(k)}$, the iterate $\hat{z}_i^{(k+1)}$ is replaced by the center of the interval with the fixed-point iterative process continuing on the subinterval containing $\hat{z}_i^*$. This bisection step additionally guarantees that either the left endpoint grows by a factor of $1 + \theta$ or the right endpoint shrinks by a factor of $1 - \theta$ in each iteration, where $\theta > 0$ is a parameter that is of smaller order than the desired level of accuracy for the fixed-point approximation.

We now have the following theoretical result establishing the convergence properties of our modified fixed-point iterative process.

Theorem 7. The iterates $\{\hat{z}_i^{(k)}\}$ of our modified fixed-point iterative process converge to the unique fixed point.

6 Numerical Experiments

In this section we present a representative sample of the computational results from a large collection of numerical experiments that support and validate our general theoretical foundation for the mathematical optimization and performance analysis of resource allocation in

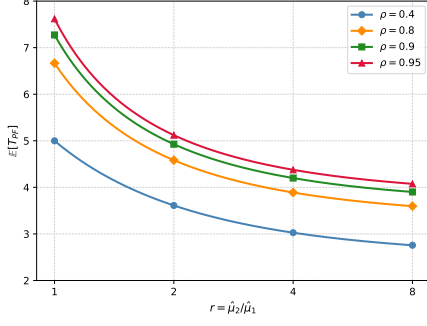
QCSC environments. We first consider numerical experiments for the two basic core components of QCSC application workloads, starting with sequential computational phases and then moving to parallel computational phases. We then focus our attention on numerical experiments for various aspects of the optimal resource capacity allocation policy, including a performance comparison with the existing reservation-based resource allocation policy. Additional numerical results and details can be found in Appendix D.

6.1 Core Sequential Computational Phases

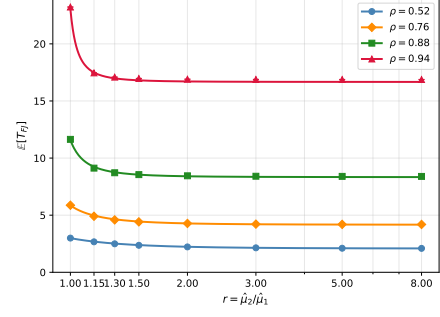
Let us consider the expected sojourn time $\mathbb{E}[T_{PF}]$ for the basic core sequential computational phase from (2) with respect to the arrival rate γ and the heterogeneity ratio r . Figure 1a plots curves for $\mathbb{E}[T_{PF}]$ as a function of $r \in \{1, 2, 4, 8\}$ under various system load conditions $\rho = \gamma/\mu^* \in \{0.4, 0.8, 0.9, 0.95\}$, where $\hat{\mu}_1 = \hat{\mu}_3 = 1$, $\hat{\mu}_2 = \hat{\mu}_4 = r$, and $p_o = 0.5$. In this case, since $r \geq 1$, we have $\mu^* = 1$ and the utilization of queues 1 and 3 and queues 2 and 4 are $\rho/2$ and $\rho/2r$, respectively. Hence, $\mathbb{E}[T_{PF}] = 4/(1 - \rho/2)$ for $r = 1$, depicted by the leftmost points in the graph. Further, as $r \rightarrow \infty$, the response time is mainly due to queues 1 and 3, i.e., $\mathbb{E}[T_{PF}] = 2/(1 - \rho/2)$, depicted by the limits of the rightmost points in the graph, approaching the value of 2 at $\rho = 0$. In general, we observe that $\mathbb{E}[T_{PF}]$ decreases monotonically with an increasing heterogeneity ratio r , where the magnitude of $\mathbb{E}[T_{PF}]$ is larger for higher load conditions under a given r as expected. Here, the faster processing units at queues 2 and 4 contribute less to the expected sojourn time as $r > 1$ grows, causing the slower processing units at queues 1 and 3 to become more and more of a bottleneck that dominates the expected sojourn time. Given the symmetry of our mathematical model of the core sequential computational phase, the bottleneck processing units at queues 1 and 3 will represent QPUs in the case of quantum-dominant application workloads and will represent GPUs in the case of classical-dominant application workloads. Moreover, due to this symmetry together with the above setting of the service rates, the curves for $\mathbb{E}[T_{PF}]$ in Figure 1a will be the same independent of $p_o \in (0, 1)$. These analytical results, in comparison and contrast with the associated results of Section 6.2, will help to guide our understanding of the optimal resource capacity allocation policy and the corresponding expected sojourn time results in Section 6.3.

6.2 Core Parallel Computational Phases

Let us now consider the expected sojourn time $\mathbb{E}[T_{FJ}]$ for the basic core parallel computational phase from our upper-lower bound interpolation in (6) and our 0th and 1st order light-heavy traffic interpolation respectively in (13) and (14). We start by examining the accuracy of our interpolation approximations for the fork-join queueing network with nonhomogeneous service rates, each of which provides the exact result in (3) for the homogeneous case when $r = 1$; refer to Remark 1 and Remark 2. Table 1 presents the expected sojourn time $\mathbb{E}[T_{FJ}]$ for each of our three approximations as a function of $r \in \{2, 4, 8\}$ under various system load conditions $\rho = \gamma/\mu^* \in \{0.4, 0.8, 0.9, 0.95\}$ together with $\mathbb{E}[T_{FJ}]$ results from a corresponding discrete-event simulation, where $\hat{\mu}_1 = 1$ and $\hat{\mu}_2 = r$. The simulation of each



(a) Core Sequential Computational Phase



(b) Core Parallel Computational Phase

Figure 1: The expected sojourn time for the two basic core components as a function of the system load $\rho = \gamma/\mu^*$ and of the heterogeneity ratio r .

instance of the fork-join queueing network consisted of 20-million jobs with a warm-up period of 500-thousand jobs repeated over five independent replicas, where the 95% confidence interval for the expected sojourn time $\mathbb{E}[T_{\text{sim}}]$ is computed via the t -distribution across the different replicas. We observe that our approximations provide highly accurate closed-form expressions for the expected sojourn time over all parameters with each error, computed as $(\mathbb{E}[T_{FJ}] - \mathbb{E}[T_{\text{sim}}])/\mathbb{E}[T_{\text{sim}}] \times 100\%$, strictly less than 1.2%. In most cases, the errors among the approximations are quite close.

Next, we consider the expected sojourn time $\mathbb{E}[T_{FJ}]$ for the basic core parallel computational phase from (6) with respect to the arrival rate γ and the heterogeneity ratio r . Figure 1b plots curves for $\mathbb{E}[T_{FJ}]$ as a function of $r \in \{1, 1.15, 1.3, 1.5, 2, 3, 5, 8\}$ under various system load conditions $\rho = \gamma/\mu^* \in \{0.52, 0.76, 0.88, 0.94\}$, where $\hat{\mu}_1 = 1$ and $\hat{\mu}_2 = r$. Here, the heterogeneity ratio is spaced more densely near $r = 1$ because $\mathbb{E}[T_{FJ}]$ transitions most steeply from the homogeneous value to the bottleneck asymptote; and the system load is based on a geometric spacing in the slack to saturation of $(1 - \rho) \in \{0.48, 0.24, 0.12, 0.06\}$. In general, we observe that the expected sojourn time $\mathbb{E}[T_{FJ}]$ sharply decreases monotonically with an increasing heterogeneity ratio r , where the magnitude of $\mathbb{E}[T_{FJ}]$ is larger for higher load conditions under a given r and even more significantly so than in Figure 1a. We further note that the faster processing units at queue 2 contribute considerably less to the expected sojourn time as $r > 1$ grows, causing the slower processing units at queue 1 to quickly become more and more of a bottleneck that dominates the expected sojourn time. In particular, the expected sojourn time $\mathbb{E}[T_{FJ}]$ curves are nearly flat across all load conditions once r increases to around 1.5. Given the symmetry of our mathematical model of the core parallel computational phase, the bottleneck processing units at queue 1 will represent QPUs in the case of quantum-dominant application workloads and will represent GPUs in the case of classical-dominant application workloads. These analytical results, in comparison and contrast with the associated results of Section 6.1, will help to guide our understanding of the optimal resource capacity allocation policy and the corresponding expected sojourn

Table 1: The expected sojourn time for the core parallel computational phase from our three interpolation approximations as a function of the system load $\rho = \gamma/\mu^*$ and a function of the heterogeneity ratio $r > 1$ in comparison against simulation, with the lowest error highlighted in bold. All three approximations are exact when $r = 1$.

ρ	r	Simulation	$\mathbb{E}[T_{FJ}]$ in (6)		$\mathbb{E}[T_{FJ}^{(0)}]$ in (13)		$\mathbb{E}[T_{FJ}^{(1)}]$ in (14)	
		$\mathbb{E}[T_{\text{sim}}] \pm 95\% \text{ CI}$	$\mathbb{E}[T]$	Err	$\mathbb{E}[T]$	Err	$\mathbb{E}[T]$	Err
0.4	2	1.816 ± 0.001	1.824	+ 0.46%	1.834	+0.97%	1.825	+0.48%
	4	1.704 ± 0.001	1.704	+ 0.00%	1.717	+0.76%	1.705	+0.04%
	8	1.677 ± 0.001	1.676	-0.10%	1.681	+0.19%	1.676	- 0.09%
0.8	2	5.109 ± 0.003	5.101	- 0.15%	5.168	+1.16%	5.151	+0.81%
	4	5.042 ± 0.003	5.016	-0.52%	5.050	+ 0.15%	5.026	-0.33%
	8	5.032 ± 0.003	5.003	-0.56%	5.014	- 0.35%	5.005	-0.53%
0.9	2	10.133 ± 0.006	10.063	-0.69%	10.170	+0.37%	10.150	+ 0.17%
	4	10.094 ± 0.006	10.009	-0.84%	10.050	- 0.43%	10.023	-0.70%
	8	10.088 ± 0.006	10.002	-0.85%	10.014	- 0.73%	10.004	-0.84%
0.95	2	20.260 ± 0.012	20.036	-1.11%	20.174	- 0.43%	20.153	-0.53%
	4	20.239 ± 0.012	20.005	-1.16%	20.050	- 0.93%	20.021	-1.07%
	8	20.236 ± 0.012	20.001	-1.16%	20.014	- 1.10%	20.003	-1.15%

time results in Section 6.3.

6.3 Optimal Resource Capacity Allocation

Finally, let us turn to consider the optimal resource capacities from the solution of (OPT_{RCA}) together with the corresponding expected sojourn time $\mathbb{E}[T]$ as a function of the arrival rate λ , heterogeneity ratio r , workload parameters p_{H} and p_{O} , cost vector $\mathbf{c}$, and budgetary cost $\mathcal{C}$. We present three primary sets of results from representative samples of numerical experiments based on our (modified) fixed-point iterative process derived in Section 5 together within the context of an expanded version of the simulation of Section 6.2. These representative samples span ranges of the key parameters of arrival rates, heterogeneity ratios, workload parameters, and cost vectors, with a larger collection of associated numerical experiments provided in Appendix D. We also present corresponding results for different budgetary costs in Appendix D.

Figure 2 presents the optimal resource capacity allocations for the quantum and classical processing units under different parameters of our QCSC application and system models; namely, the cumulative optimal resource capacity allocations for system QPUs along the x-axis and the cumulative optimal resource capacity allocations for system GPUs along the y-axis, both with respect to the different model parameters. We observe that the optimal allocation is essentially equal for unit costs and when $r = 1$ (top row). As r increases, however, the bottleneck servers of the core parallel and sequential computational phases

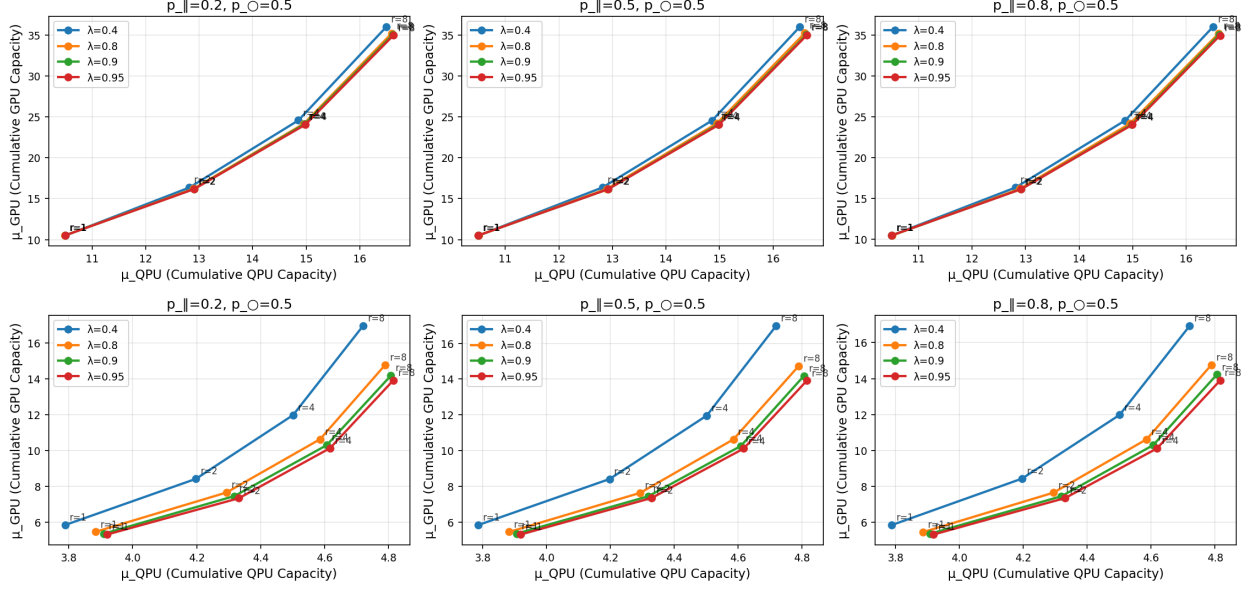


Figure 2: The cumulative optimal resource capacity allocations for system QPUs and GPUs under different parameters of our QCSC models. The y-axis depicts the cumulative GPU capacities and the x-axis depicts the cumulative QPU capacities for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$ and $r \in \{1, 2, 4, 8\}$, under unit costs for QPUs and GPUs in the top row and under costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom row. Both rows fix $p_o = 0.5$ while varying $p_{||} \in \{0.2, 0.5, 0.8\}$.

receive a greater resource capacity allocation, corresponding in this specific case to the GPUs. We note that, given certain symmetries of our QCSC models, the same trend can be observed with the QPUs receiving a greater resource capacity allocation in the specific case of QPUs as the bottleneck servers of the core parallel and sequential computational phases. These performance trends are consistent across all system loads, with somewhat higher allocations for the bottleneck servers under lighter loads. Under costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs, there is a decrease in the optimal resource capacity allocations overall due to the fixed cost budget $\mathcal{C}$; in addition, the bottleneck servers (GPUs) of the core parallel and sequential computational phases receive an even greater proportion of the resource capacity allocation due to the larger costs for the QPUs. Once again, given certain symmetries of our QCSC models, the same trend can be observed with the QPUs receiving the even greater proportion of the resource capacity allocation in the case of switched costs and the QPUs as the bottleneck servers of the core parallel and sequential computational phases. These performance trends are fairly consistent across all heavier system loads, with higher allocations for the bottleneck servers (GPUs) under lighter loads. We note that the trends in Figure 2 for fixed p_o and varying $p_{||}$ are the same as those in Figure 6 for fixed $p_{||}$ and varying p_o , due to certain symmetries of our QCSC models.

Figure 3 presents the expected sojourn time under the optimal resource capacity allocations for the quantum and classical processing units as a function of the heterogeneity ratio

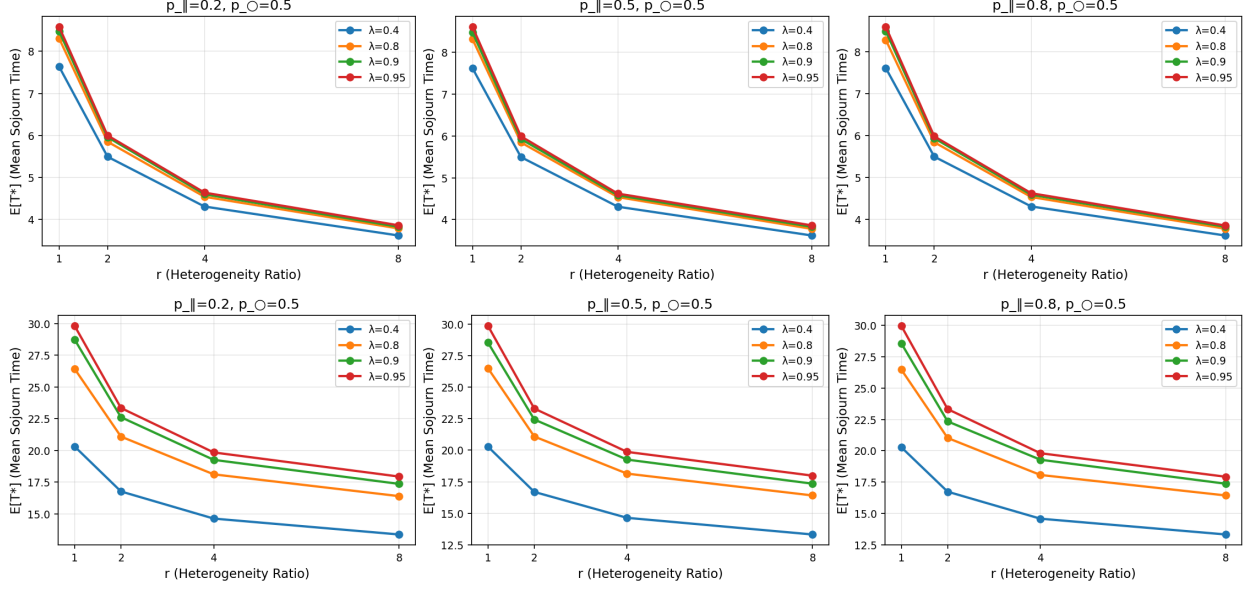


Figure 3: The expected sojourn time under the optimal resource capacity allocations as a function of the heterogeneity ratio $r \in \{1, 2, 4, 8\}$ for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$, with unit costs for QPUs and GPUs in the top row and with costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom row. Both rows fix $p_o = 0.5$ while varying $p_{||} \in \{0.2, 0.5, 0.8\}$.

for different parameters of our QCSC application and system models. We observe that the expected sojourn time is highest when $r = 1$ under both cost scenarios. As r increases, however, the greater resource capacity allocation to the bottleneck servers of the core parallel and sequential computational phases yields a decreasing trend in the expected sojourn time. We note that, given certain symmetries of our QCSC models, such trends persist irrespective of whether the GPUs or the QPUs are the bottleneck servers of the core parallel and sequential computational phases. These performance trends are consistent across all system loads, with somewhat lower expected sojourn times due to somewhat higher allocations for the bottleneck servers under the lighter system loads. Under costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs, there is a significant increase in the expected sojourn times across the board because of the combination of the decrease in optimal resource capacity allocations overall due to the fixed cost budget $\mathcal{C}$ and the even greater proportion of the resource capacity allocation to the bottleneck servers of the core parallel and sequential computational phases due to the larger costs for the QPUs. Once again, given certain symmetries of our QCSC models, such trends persist irrespective of whether the GPUs or the QPUs are the bottleneck servers of the core parallel and sequential computational phases. Yet again, these performance trends are fairly consistent across all system loads, particularly heavier loads and with higher allocations for the bottleneck servers under lighter loads rendering smaller expected sojourn times. We note that the trends in Figure 3 for fixed p_o and varying $p_{||}$ are the same as those in Figure 7 for fixed $p_{||}$ and varying p_o , due to certain symmetries of our QCSC models.

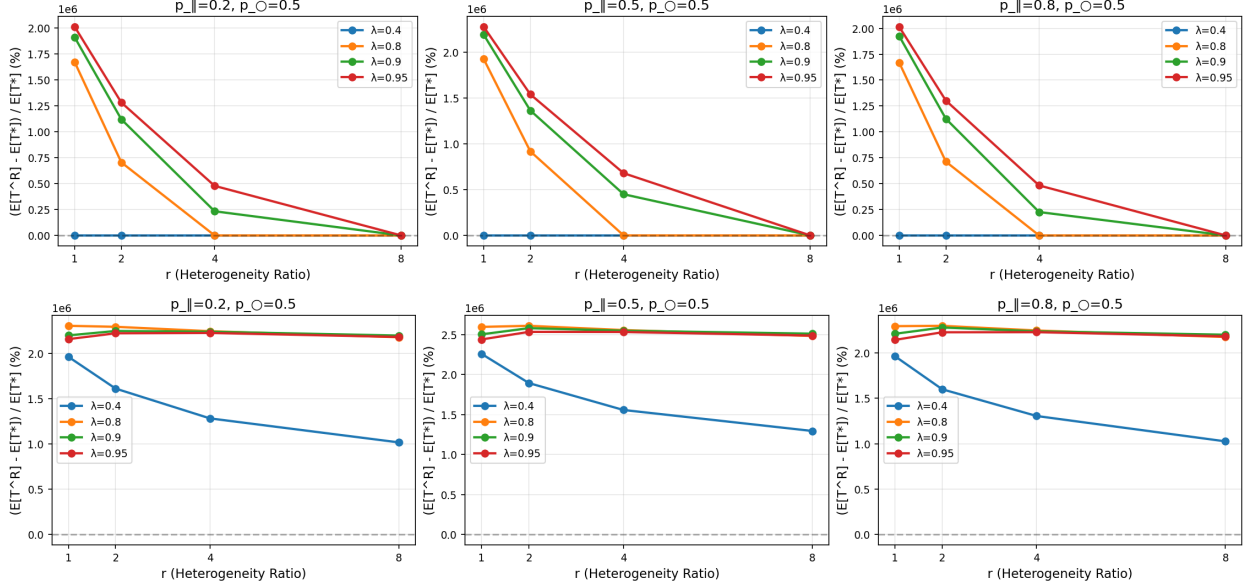


Figure 4: The relative improvement in expected sojourn time under the optimal resource capacity allocations over that under the reservation-based resource allocation policy as a function of the heterogeneity ratio $r \in \{1, 2, 4, 8\}$ for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$, with unit costs for QPUs and GPUs in the top row and with costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom row. Both rows fix $p_\circ = 0.5$ while varying $p_\parallel \in \{0.2, 0.5, 0.8\}$.

Figure 4 presents the relative expected sojourn-time improvement from the optimal resource capacity allocations for the quantum and classical processing units in comparison with the reservation-based resource allocation policy of Section 4. Namely, the curves depict $(E[T^R] - E[T^*]) / E[T^*]$ as a function of the heterogeneity ratio for different parameters of our QCSC application and system models, where $E[T^*]$ is the expected sojourn time under the optimal resource capacity allocations, and $E[T^R]$ is the expected sojourn time under the reservation-based policy with the same optimal resource capacities. We note that, given the scale of the y-axis, the optimal resource capacity allocation policy provides significant expected sojourn-time improvements over the reservation-based resource allocation policy in all cases, especially under heavier system loads. These relative performance improvements are highest when $r = 1$. As r increases, however, the greater resource capacity allocation to the bottleneck servers of the core parallel and sequential computational phases yields a decreasing trend in the relative expected sojourn-time improvements. We further note that, given certain symmetries of our QCSC models, such trends persist irrespective of whether the GPUs or the QPUs are the bottleneck servers of the core parallel and sequential computational phases. These performance trends are again consistent across all system loads, with somewhat lower expected sojourn times due to the lighter loads and somewhat higher allocations for the bottleneck servers. Under costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs, there is a significant increase in the expected sojourn times across the board for the reasons noted above, and these effects have an even greater impact under the reservation-based

policy. This in turn results in smaller decreasing trends in the relative performance improvements as r increases. Once again, given certain symmetries of our QCSC models, such trends persist irrespective of whether the GPUs or the QPUs are the bottleneck servers of the core parallel and sequential computational phases. Yet again, these performance trends are fairly consistent across all system loads, particularly heavier loads and with higher allocations for the bottleneck servers under lighter loads rendering smaller expected sojourn times. We note that the trends in Figure 4 for fixed p_o and varying p_{II} are the same as those in Figure 8 for fixed p_{II} and varying p_o , due to the symmetries of our QCSC models.

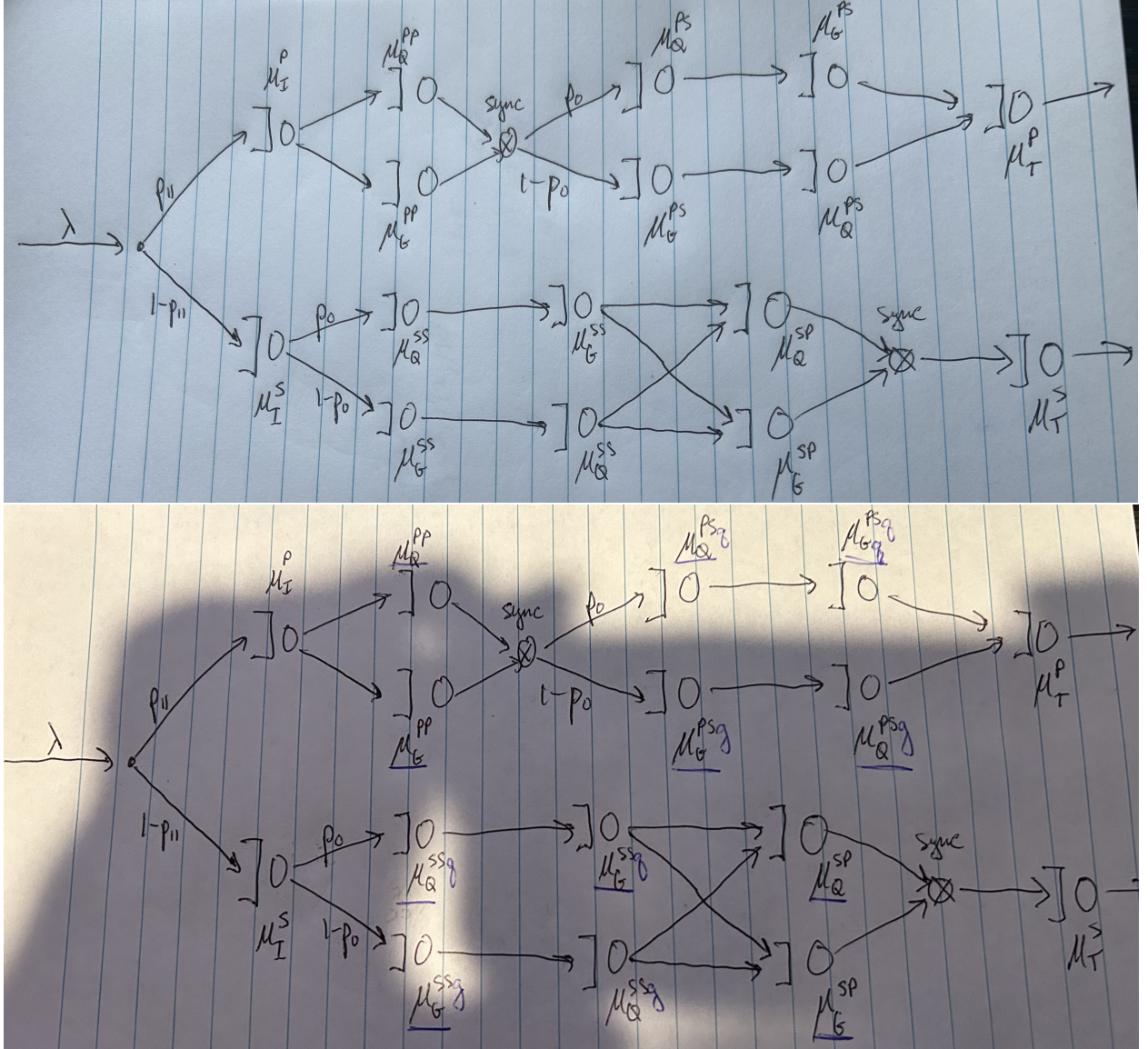


Figure 5: Illustration of Quantum-Centric Supercomputing Environment. This is solely intended to illustrate the queueing network that we want to simulate.

7 Conclusions

We consider in this paper the problem of determining the optimal allocation of resource capacities in quantum-centric supercomputing environments. In strong contrast with traditional high-performance computing environments, the system architectures of such environments comprise combinations of quantum and classical processing units and the application workloads of such environments comprise computation patterns spanning quantum and classical processing units. Our focus is on deriving a general theoretical foundation for the mathematical optimization and performance analysis of resource allocation in quantum-centric supercomputing environments. This includes a mathematical analysis of basic computational phases at the core of application workload patterns, leading to exact closed-form solutions for the sequential computational phase and closed-form approximations for the parallel computational phase, and a mathematical analysis of fairly common reservation-based resource allocation policies. We then derive an efficient numerical solution of the general resource capacity allocation optimization problem that establishes an optimal resource allocation policy, that establishes theoretical properties on the guaranteed convergence of the corresponding iterative process to a unique fixed point, and that provides fundamental insights into resource allocation in quantum-centric supercomputing environments. Numerical experiments demonstrate and quantify the significant performance benefits of our optimal allocation policy over the existing reservation-based allocation policy across different application workloads. This also includes evaluating the performance of the basic core computational phases across different workloads and demonstrating the very high accuracy of our closed-form approximations for the expected sojourn time of the corresponding fork-join queueing networks under heterogeneous service rates, of interest in their own right.

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A Proofs of Main Theoretical Results

In this section, we provide the proofs of our main theoretical results throughout the paper, with additional details and more complete proofs provided in the appendices.

A.1 Proof of Lemma 1

The sojourn times $T_{FJ}^{(1)}$ and $T_{FJ}^{(2)}$ of the first and second tasks of arriving jobs share a common Poisson arrival process, and thus they are positively correlated. It is well known that positive correlation reduces the expected maximum $\mathbb{E}[\max\{T_{FJ}^{(1)}, T_{FJ}^{(2)}\}]$ below what it would be if $T_{FJ}^{(1)}$ and $T_{FJ}^{(2)}$ were independent [5]. Hence, upon considering $T_{FJ}^{(1)}$ and $T_{FJ}^{(2)}$ as independent exponential random variables, respectively with rates $\hat{\mu}_1 - \gamma$ and $\hat{\mu}_2 - \gamma$, representing the sojourn time of independent M/M/1 queues, we obtain the upper bound in (4) due to the known relationship $\mathbb{E}[\max\{Z_1, Z_2\}] = \mathbb{E}[Z_1] + \mathbb{E}[Z_2] - \mathbb{E}[\min\{Z_1, Z_2\}]$ for independent exponential random variables Z_1 and Z_2 .

Turning to the lower bound, consider the expected sojourn time of the M/M/1 queue with arrival rate γ and service rate $\hat{\mu}_i$ which is given by $\mathbb{E}[T_{MM1}^{(i)}] = 1/(\hat{\mu}_i - \gamma)$, $i = 1, 2$. It follows from Jensen's inequality and the convexity of the maximum function that the expected sojourn time of the fork-join queueing network $\mathbb{E}[T_{FJ}]$ is bounded below by $\max\{\mathbb{E}[T_{MM1}^{(1)}], \mathbb{E}[T_{MM1}^{(2)}]\}$, which represents the expected sojourn time of the bottleneck M/M/1 queue and yields the lower bound in (5).

A.2 Proof of Theorem 1

The upper-lower bound interpolation approximation is based on a convex combination of the upper and lower bounds. Specifically, consider the convex combination

$$\mathbb{E}[T_{FJ}] = (1 - \alpha)\mathbb{E}[T_{FJ}^{UB}] + \alpha\mathbb{E}[T_{FJ}^{LB}] \quad (32)$$

where we seek to determine α from the known exact result for homogeneous service rates. Let $\mathbb{E}[T_{MM1}]$ denote the expected sojourn time of the M/M/1 queue with arrival rate γ and service rate $\hat{\mu}$. When $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$, we have $\mathbb{E}[T_{FJ}^{UB}] = (3/2) \cdot \mathbb{E}[T_{MM1}]$ and $\mathbb{E}[T_{FJ}^{LB}] = \mathbb{E}[T_{MM1}]$, which yields the following relationship between the exact result for $\mathbb{E}[T_{FJ}]$ in (3) and the convex combination for $\mathbb{E}[T_{FJ}]$ in (32) with respect to α :

$$\frac{12 - \gamma/\hat{\mu}}{8(\hat{\mu} - \gamma)} = (1 - \alpha)(3/2)\mathbb{E}[T_{MM1}] + \alpha\mathbb{E}[T_{MM1}] = \frac{(1 - \alpha)(3/2)}{\hat{\mu} - \gamma} + \frac{\alpha}{\hat{\mu} - \gamma}.$$

Then, upon solving for α , we obtain

$$\alpha = 3 - \frac{12 - \gamma/\hat{\mu}}{4} = \frac{\gamma/\hat{\mu}}{4}.$$

For the fork-join queueing network of interest with nonhomogeneous rates, a natural generalization is to replace $\gamma/\hat{\mu}$ above with the average utilization $(\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)/2$, thus yielding

$$\hat{\alpha} = \frac{(\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)/2}{4} = \frac{(\gamma/\hat{\mu}_1 + \gamma/\hat{\mu}_2)}{8}.$$

Upon substituting (4) and (5) together with the above expression for $\hat{\alpha}$ into (32), we have the desired result in (6), thus completing the proof.

A.3 Proof of Lemma 2

Since the tagged job does not encounter any prior arrivals, we have by definition that $\mathbb{E}[\psi(\emptyset)] = \mathbb{E}[\max\{v_1, v_2\}]$ where the random variable $\psi(\emptyset) = \max\{v_1, v_2\}$ follows a phase-type distribution $(\alpha, \mathbf{Q}_{FJ})$ [34, 47, 49] defined on a four-state Markov process with infinitesimal generator matrix $\mathbf{Q}_{FJ}$ expressed as

$$\mathbf{Q}_{FJ} = \begin{bmatrix} \tilde{\mathbf{Q}} & \tilde{\mathbf{q}} \\ \mathbf{0} & 0 \end{bmatrix}, \quad \tilde{\mathbf{Q}} = \begin{bmatrix} -(\hat{\mu}_1 + \hat{\mu}_2) & \hat{\mu}_1 & \hat{\mu}_2 \\ 0 & -\hat{\mu}_2 & 0 \\ 0 & 0 & -\hat{\mu}_1 \end{bmatrix}, \quad \tilde{\mathbf{q}} = \begin{bmatrix} 0 \\ \hat{\mu}_2 \\ \hat{\mu}_1 \end{bmatrix}, \quad (33)$$

together with the initial probability vector $(\boldsymbol{\alpha}, \alpha_{00})$ where $\boldsymbol{\alpha} = [1, 0, 0]$ and $\alpha_{00} = 0$. Defining $\exp(\mathbf{Y}) := \sum_{j=0}^{\infty} \frac{1}{j!} \mathbf{Y}^j$ for any given matrix $\mathbf{Y}$, the distribution function $H(\cdot)$ of the random variable $\psi(\emptyset)$ is given by

$$H(y) = 1 - \boldsymbol{\alpha} \exp(\tilde{\mathbf{Q}}y) \mathbf{e}, \quad y \in \mathbb{R}_{\geq 0},$$

from which we have $\mathbb{E}[\psi(\emptyset)] = -\boldsymbol{\alpha} \tilde{\mathbf{Q}}^{-1} \mathbf{e}$. Then we obtain from (33) and direct calculation

$$\tilde{\mathbf{Q}}^{-1} = \begin{bmatrix} -\frac{1}{\hat{\mu}_1 + \hat{\mu}_2} & -\frac{\hat{\mu}_1}{\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} & -\frac{\hat{\mu}_2}{\hat{\mu}_1(\hat{\mu}_1 + \hat{\mu}_2)} \\ 0 & -\frac{1}{\hat{\mu}_2} & 0 \\ 0 & 0 & -\frac{1}{\hat{\mu}_1} \end{bmatrix},$$

which together with $-\boldsymbol{\alpha} \tilde{\mathbf{Q}}^{-1} \mathbf{e}$ yields the desired result (8).

A.4 Proof of Lemma 3

We have by definition that

$$\psi(\{t\}) = \begin{cases} \max\{v_1, v_2\}, & t \in [0, \infty), \\ \max\{v_1 + (w_1 + t)^+, v_2 + (w_2 + t)^+\}, & t \in (-\infty, 0), \end{cases}$$

where $(x)^+ := \max\{0, x\}$, from which together with (9) we conclude

$$f^{(1)}(0) = \int_{-\infty}^0 (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt, \quad (34)$$

given that $\mathbb{E}[\psi(\{t\})] = \mathbb{E}[\max\{v_1, v_2\}] = \mathbb{E}[\psi(\emptyset)]$ for $t \in [0, \infty)$. By definition, the random variables $v_1 + (w_1 + t)^+$ and $v_2 + (w_2 + t)^+$ for fixed $t \in (-\infty, 0)$ are independent and respectively follow the distribution function given by

$$\begin{aligned} H_1(y) &= 1 - \hat{H}_1(y), & H_2(y) &= 1 - \hat{H}_2(y), & y &\in \mathbb{R}_{\geq 0}, \\ \hat{H}_1(y) &= (1 + \hat{\mu}_1 y \exp(\hat{\mu}_1 t)) \exp(-\hat{\mu}_1 y), & \hat{H}_2(y) &= (1 + \hat{\mu}_2 y \exp(\hat{\mu}_2 t)) \exp(-\hat{\mu}_2 y). \end{aligned}$$

From the independence of these two random variables, we obtain for fixed $t \in (-\infty, 0)$

$$\begin{aligned} \mathbb{E}[\psi(\{t\})] &= \int_0^{\infty} (1 - H_1(y) H_2(y)) dy \\ &= \int_0^{\infty} \hat{H}_1(y) dy + \int_0^{\infty} \hat{H}_2(y) dy - \int_0^{\infty} \hat{H}_1(y) \hat{H}_2(y) dy, \\ &= \frac{1 + e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{1 + e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3}. \end{aligned} \quad (35)$$

Upon substituting (35) and (8) from Lemma 2 into (34), we then derive

$$\begin{aligned}
f^{(1)}(0) &= \int_{-\infty}^0 \left(\frac{e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3} + \frac{1}{\hat{\mu}_1} + \frac{1}{\hat{\mu}_2} - \frac{1}{\hat{\mu}_1 + \hat{\mu}_2} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)} \right) dt \\
&= \int_{-\infty}^0 \left(\frac{e^{\hat{\mu}_1 t}}{\hat{\mu}_1} + \frac{e^{\hat{\mu}_2 t}}{\hat{\mu}_2} - \frac{\hat{\mu}_1 e^{\hat{\mu}_1 t} + \hat{\mu}_2 e^{\hat{\mu}_2 t}}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2 e^{(\hat{\mu}_1 + \hat{\mu}_2)t}}{(\hat{\mu}_1 + \hat{\mu}_2)^3} \right) dt, \\
&= \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1 \hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4},
\end{aligned}$$

thus yielding the desired result.

A.5 Proof of Lemma 4

Taking the limit as $r \rightarrow 1$, the application workload of the fork-join queueing network becomes balanced. The heavy-traffic limit for the fork-join queueing network with homogeneous service rates is known to be $h = 11/8$ [68], thus establishing the upper bound in (50). In the other direction, taking the limit as $r \rightarrow \infty$, the application workload of the fork-join queueing network becomes unbalanced to the point of reducing the stochastic network to the corresponding M/M/1 queue. The heavy-traffic limit for the M/M/1 queue is known to be $h = 1$ [24, 71], thus establishing the lower bound in (50).

Our upper-lower bound approximation smoothly interpolates between the upper-bound (homogeneous) limit of $11/8$ and the lower-bound (bottleneck) limit of 1 according to the power-law function in (12) with respect to the heterogeneity ratio r . The interpolation approximation satisfies $h(1) = 11/8$ for all values of β , thus matching the exact result for $r = 1$. The interpolation approximation further satisfies the lower bound $h(r) \rightarrow 1$ as $r \rightarrow \infty$, with β controlling the rate of decay from the upper bound toward the lower bound with respect to the heterogeneity ratio r .

A.6 Proof of Theorem 2

We start with the 0^{th} order light-heavy traffic interpolation approximation $\tilde{f}_0(\gamma)$ in (13) for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1)$, based on a combination of the 0^{th} order light-traffic derivative and the heavy-traffic limit. The combination of (42), (44), and (45) in Appendix B yields

$$g(\gamma) = (\hat{\mu}_1 - \gamma) (f^{(0)}(0) + \gamma h),$$

from which together with (8) in Lemma 2 and (12) in Lemma 4 we derive

$$\begin{aligned}
g(0) &= (\hat{\mu}_1 - 0) \left(\frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)} + 0 \cdot f^{(1)}(0) + 0 \cdot h \right) = \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)}, \\
g(\mu^*) &= g(\hat{\mu}_1) = h^* = h(r).
\end{aligned}$$

With respect to (43) in Appendix B, the linear interpolation $\tilde{g}(\gamma)$ corresponding to $g(\gamma)$ is given by

$$\tilde{g}(\gamma) = a_1 \gamma + a_0.$$

From the matching light-traffic conditions (46) and the heavy-traffic condition (47) in Appendix B we have

$$\begin{aligned} a_1 &= \frac{1}{\hat{\mu}_1}(h(r) - a_0) = \frac{1}{\hat{\mu}_1}h(r) - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \\ a_0 &= \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \end{aligned}$$

from which we obtain

$$\begin{aligned} \tilde{g}(\gamma) &= \gamma \left(\frac{1}{\hat{\mu}_1}h(r) - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \right) + \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\ &= \frac{\gamma}{\hat{\mu}_1}h(r) + (\hat{\mu}_1 - \gamma) \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \end{aligned}$$

Then, given this linear interpolation $\tilde{g}(\gamma)$, we obtain from (48) in Appendix B the desired 0^{th} order interpolation approximation $\tilde{f}_0(\gamma)$ of $f(\gamma)$ as

$$\tilde{f}_0(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \frac{h(r)}{\hat{\mu}_1} + \frac{\hat{\mu}_1 - \gamma}{\hat{\mu}_1 - \gamma} \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \quad (36)$$

Next, we turn to the 1^{st} order light-heavy traffic interpolation approximation $\tilde{f}_1(\gamma)$ in (14) for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1)$, based on a combination of the 0^{th} order and 1^{st} order light-traffic derivatives and the heavy-traffic limit. The combination of (42), (44), and (45) in Appendix B yields

$$g(\gamma) = (\hat{\mu}_1 - \gamma) \left(f^{(0)}(0) + \gamma f^{(1)}(0) + \frac{\gamma^2}{2}h \right),$$

from which together with (8) in Lemma 2, (10) in Lemma 3, and (12) in Lemma 4 we derive

$$\begin{aligned} g(0) &= (\hat{\mu}_1 - 0) \left(\frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} + 0 \cdot f^{(1)}(0) + 0 \cdot h \right) = \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\ g'(0) &= \frac{d}{d\gamma} \left[(\hat{\mu}_1 - \gamma) \left(f^{(0)}(0) + \gamma f^{(1)}(0) + \frac{\gamma^2}{2}h \right) \right] \Big|_{\gamma=0}, \\ &= \frac{d}{d\gamma} \left[\hat{\mu}_1 f^{(0)}(0) + \gamma \hat{\mu}_1 f^{(1)}(0) + \frac{\gamma^2 \hat{\mu}_1}{2}h - \gamma f^{(0)}(0) - \gamma^2 f^{(1)}(0) - \frac{\gamma^3}{2}h \right] \Big|_{\gamma=0}, \\ &= \left[0 + \hat{\mu}_1 f^{(1)}(0) + \gamma \hat{\mu}_1 h - f^{(0)}(0) - 2\gamma f^{(1)}(0) - \frac{3\gamma^2}{2}h \right] \Big|_{\gamma=0}, \\ &= \hat{\mu}_1 f^{(1)}(0) - f^{(0)}(0) = \frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\ g(\mu^*) &= g(\hat{\mu}_1) = h^*. \end{aligned}$$

With respect to (43) in Appendix B, the quadratic interpolation $\tilde{g}(\gamma)$ corresponding to $g(\gamma)$ is given by

$$\tilde{g}(\gamma) = a_2\gamma^2 + a_1\gamma + a_0.$$

From the matching light-traffic conditions (46) and the heavy-traffic condition (47) in Appendix B we have

$$\begin{aligned} a_2 &= 0, \\ a_1 &= \frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\ a_0 &= \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \end{aligned}$$

from which we obtain

$$\begin{aligned} \tilde{g}(\gamma) &= \gamma \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} - \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)} \right) + \hat{\mu}_1 \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \\ &= \gamma \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + (\hat{\mu}_1 - \gamma) \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \end{aligned}$$

Then, given this quadratic interpolation $\tilde{g}(\gamma)$, we obtain from (48) in Appendix B the desired 1st order interpolation approximation $\tilde{f}_1(\gamma)$ of $f(\gamma)$ as

$$\tilde{f}_1(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1 - \gamma}{\hat{\mu}_1 - \gamma} \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}, \quad (37)$$

for $\gamma \in [0, \hat{\mu}_1)$, thus completing the proof.

A.7 Proof of Theorem 5

The Lagrangian associated with $(\text{OPT}_{\text{RCA:FF}})$ is defined as

$$\sum_{i=1}^N \frac{\omega_i \zeta_i}{\mathcal{S}_i \hat{\mu}_i - \gamma_i} + \chi \left(\sum_{i=1}^N c_i \mathcal{S}_i - \mathcal{C} \right), \quad (38)$$

where χ is the Lagrange multiplier. Taking the derivative of (38) with respect to $\mathcal{S}_i$ yields

$$\chi c_i - \frac{\omega_i \zeta_i \hat{\mu}_i}{(\mathcal{S}_i \hat{\mu}_i - \gamma_i)^2},$$

from which upon setting the result to zero and solving for $\mathcal{S}_i$ we obtain

$$\mathcal{S}_i = \frac{\gamma_i}{\hat{\mu}_i} + \frac{1}{\sqrt{\chi}} \sqrt{\frac{\omega_i \zeta_i}{c_i \hat{\mu}_i}}. \quad (39)$$

Substituting (39) into the constraint of (19) set to equality then renders

$$\sum_{i=1}^N c_i \mathcal{S}_i = \sum_{i=1}^N c_i \frac{\gamma_i}{\hat{\mu}_i} + \frac{1}{\sqrt{\chi}} \sum_{i=1}^N \sqrt{\frac{\omega_i \zeta_i c_i}{\hat{\mu}_i}} = \mathcal{C},$$

from which we have

$$\frac{1}{\sqrt{\chi}} = \frac{\mathcal{C} - \sum_{i=1}^N c_i \gamma_i / \hat{\mu}_i}{\sum_{i=1}^N \sqrt{\omega_i \zeta_i c_i / \hat{\mu}_i}}. \quad (40)$$

Upon substituting (40) into (39), we obtain the desired result

$$\mathcal{S}_i^* = \frac{\gamma_i}{\hat{\mu}_i} + \left(\mathcal{C} - \sum_{j=1}^N c_j \frac{\gamma_j}{\hat{\mu}_j} \right) \frac{\sqrt{\omega_i \zeta_i / (c_i \hat{\mu}_i)}}{\sum_{j=1}^N \sqrt{\omega_j \zeta_j c_j / \hat{\mu}_j}}.$$

A.8 Proof of Lemma 5

We treat each of the four properties of the lemma in turn.

The steady-state expected sojourn time at queue i under the Brownian approximation of our stochastic queueing network with per-queue functional form is proportional to $1/(\mathcal{S}_i \hat{\mu}_i - \gamma_i)$, $i \in [\hat{N}]$, which is convex in $\mathcal{S}_i \hat{\mu}_i$ under the stability criteria of $\mathcal{S}_i \hat{\mu}_i > \gamma_i$. Given the feedforward structure of the stochastic network topology, the sojourn time at queue i is a function of all upstream resource capacity service rates, $i \in [\hat{N}]$, but the corresponding dependence consists of a composition of convex functions. Since the composition of convex and nondecreasing functions is convex, we have that $\kappa_i(\mathbf{z}_i)$ is convex in $\mathbf{z}_i$, $i \in [\hat{N}]$.

Upon increasing any resource capacity service rate $\mathcal{S}_j \hat{\mu}_j$, $j \leq i$, the expected sojourn time at queue i decreases due to the reduced effective load at queue i , for $j, i \in [\hat{N}]$. The drift vector of the corresponding reflected Brownian motion in the positive orthant is determined by the difference between the resource capacity service rates and the effective arrival rates, with the covariance structure determined by the variability in the arrival and service times where the exponential service time variance equals the associated mean.. Hence, increasing any resource capacity service rate under the Brownian approximation increases the drift away from the boundary, thus decreasing the steady-state expected sojourn time. It therefore follows that $\kappa_i(\cdot)$ is nonincreasing in each coordinate.

For the last coordinate of $\kappa_i(\cdot)$, increasing the corresponding resource capacity service rate $\mathcal{S}_i \hat{\mu}_i$ strictly increases the drift at queue i , $i \in [\hat{N}]$, and thus strictly decreases the steady-state expected sojourn time. Hence, $\kappa_i(\cdot)$ is strictly decreasing in the last coordinate $\mathcal{S}_i \hat{\mu}_i$.

Finally, the homogeneity property follows from the scaling property of the Brownian motion and the linear dependence of the drift and covariance on the resource capacity service rates in the feedforward Jackson network. More specifically, let $T(\mathbf{z}_i)$ denote the steady-state sojourn time at queue i for parameter $\mathbf{z}_i$, $i \in [\hat{N}]$. Under the Brownian approximation, $T(\delta \mathbf{z}_i)$ is equal in distribution to $\delta T(\mathbf{z}_i)$, from which we have

$$\kappa_i(\delta \mathbf{z}_i) = \mathbb{E}[T(\delta \mathbf{z}_i)] = \mathbb{E}[\delta T(\mathbf{z}_i)] = \delta \mathbb{E}[T(\mathbf{z}_i)] = \delta \kappa_i(\mathbf{z}_i),$$

thus completing the proof.

A.9 Proof of Lemma 6

Since $\mathcal{S}_i \hat{\mu}_i = \gamma_i + \hat{z}_i(\mathcal{S}_{\varpi(i)} \hat{\mu}_{\varpi(i)} - \gamma_{\varpi(i)})$ by definition, we have $\sum_{i=1}^{\hat{N}} c_i \mathcal{S}_i = \sum_{i=1}^{\hat{N}} c_i \gamma_i / \hat{\mu}_i + \sum_{i=1}^{\hat{N}} (\mathcal{S}_{\varpi(i)} \hat{\mu}_{\varpi(i)} - \gamma_{\varpi(i)}) c_i \hat{z}_i / \hat{\mu}_i$. Upon setting the right-hand side to $\mathcal{C}$, we obtain $\mathcal{S}_1 \hat{\mu}_1, \mathcal{S}_2 \hat{\mu}_2$ which therefore fixes $\mathcal{S}_i \hat{\mu}_i$ via $\hat{z}_i = (\mathcal{S}_i \hat{\mu}_i - \gamma_i) / (\mathcal{S}_{\varpi(i)} \hat{\mu}_{\varpi(i)} - \gamma_{\varpi(i)})$ for $i \in [\hat{N}]^-$.

A.10 Proof of Theorem 6

By establishing that there is exactly one solution to the equations

$$\kappa_i(\hat{\mathbf{z}}_i) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}) = \zeta_{\varpi(i)} \frac{c_i / \omega_i}{c_{\varpi(i)} / \omega_{\varpi(i)}} \hat{z}_i, \quad i \in [\hat{N}], \quad (41)$$

the desired result then follows in light of Lemma 6. To this end, observe that all equations corresponding to queues that are a distance of one from either root have the form $\kappa_i(1, \hat{z}_i) = p_{1,i} \zeta_1 + \zeta_1 (c_i / \omega_i) / (c_1 / \omega_1) \hat{z}_i$ or $\kappa_i(2, \hat{z}_i) = p_{2,i} \zeta_2 + \zeta_2 (c_i / \omega_i) / (c_2 / \omega_2) \hat{z}_i$, depending upon the root vertex of the queues. From Lemma 5, each left-hand side is continuous and nonincreasing in $\hat{z}_i > 0$ from $+\infty$ while each right-hand side is strictly increasing in $\hat{z}_i$, and thus there is exactly one $\hat{z}_i$ -value for which equality holds. Assuming that the $\hat{z}_j$ corresponding to queues which are a distance of $m-1$ from either root have been determined from (41), now consider a queue i that is a distance of m from its root. Once again, the left-hand side of (41) is nonincreasing in $\hat{z}_i$ while the right-hand side is strictly increasing in $\hat{z}_i$, and thus there is a unique $\hat{z}_i$ for which equality holds.

A.11 Proof of Lemma 7

Let $\tau_i = c_{\varpi(i)} \omega_i / (c_i \omega_{\varpi(i)} \zeta_{\varpi(i)})$. From Lemma 5, under the first supposition $\hat{z}_i^{(k)} < \hat{z}_i^*$, we have

$$\kappa_i(\varpi(i), \hat{z}_i^{(k)}) - p_{\varpi(i),i} \zeta_{\varpi(i)} \geq \kappa_i(\varpi(i), \hat{z}_i^*) - p_{\varpi(i),i} \zeta_{\varpi(i)} = \tau_i^{-1} \hat{z}_i^*,$$

from which together with (27) and (29) we obtain

$$\hat{z}_i^{(k+1)} = \sqrt{\tau_i \hat{z}_i^{(k)} (\kappa_i(\varpi(i), \hat{z}_i^{(k)}) - p_{\varpi(i),i} \zeta_{\varpi(i)})} \geq \sqrt{\hat{z}_i^* \hat{z}_i^{(k)}} > \hat{z}_i^{(k)}.$$

Similar arguments, under the second supposition $\hat{z}_i^{(k)} > \hat{z}_i^*$, leads to $\hat{z}_i^{(k+1)} \leq \sqrt{\hat{z}_i^* \hat{z}_i^{(k)}} < \hat{z}_i^{(k)}$.

A.12 Proof of Lemma 8

By assumption, $\hat{z}_j^{(k)} \in [\hat{z}_j^* \pm \varepsilon]$ for sufficiently large k , all queues $j \in \mathcal{P}(\iota(i))$, and any $\varepsilon > 0$. Let $\tau_i = c_{\varpi(i)} \omega_i / (c_i \omega_{\varpi(i)} \zeta_{\varpi(i)})$. Since each $\kappa_j(\cdot)$ is a continuous function on $(0, \infty)^j$ by convexity, the contour $\{\hat{\mathbf{z}}_i \in (0, \infty)^{|\mathcal{P}(i)|} : \kappa_i(\hat{\mathbf{z}}_i) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}) = \tau_i^{-1} \hat{z}_i^*\}$ is continuous in any neighborhood of $\hat{\mathbf{z}}_i^*$. From Lemma 5, $\kappa_i(\hat{\mathbf{z}}_i) - p_{\iota(i),i} \kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)})$ is strictly increasing in $\hat{z}_i$, and therefore it follows from a suitable version of the implicit function theorem that

there exists a continuous function $h_i : (0, \infty)^{|\mathcal{P}(i)|-1} \rightarrow (0, \infty)$ such that the above contour coincides with the graph $\{(\hat{\mathbf{z}}_{\iota(i)}, h_i(\hat{\mathbf{z}}_{\iota(i)})) : \hat{\mathbf{z}}_{\iota(i)} \in (0, \infty)^{|\mathcal{P}(i)|-1} \text{ with } h_i \text{ defined as}$

$$\begin{aligned} h_i(\hat{\mathbf{z}}_{\iota(i)}) &= \inf\{\hat{z}_i > 0 : \kappa_i(\hat{\mathbf{z}}_i) - p_{\iota(i),i}\kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}) = \tau_i^{-1}\hat{\mathbf{z}}_i^*\} \\ &= \sup\{\hat{z}_i > 0 : \kappa_i(\hat{\mathbf{z}}_i) - p_{\iota(i),i}\kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}) = \tau_i^{-1}\hat{\mathbf{z}}_i^*\}. \end{aligned}$$

By the continuity of h_i , we can find for arbitrary $\xi > 0$ some $\delta(\xi)$ such that the range of h_i on $[\hat{z}_{\varpi(i)}^* \pm \delta(\xi)] \times \cdots \times [\hat{z}_{\iota(i)}^* \pm \delta(\xi)]$ is included in $[\hat{z}_i^* \pm \xi]$. Without loss of generality, assume that $\delta(\xi)$ is strictly increasing in ξ .

Next, we establish that the iterates move toward in the desired direction. By construction, for $\hat{z}_i > h_i(\hat{\mathbf{z}}_{\iota(i)})$, we have $\kappa_i(\hat{\mathbf{z}}_i) - p_{\iota(i),i}\kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}) < \tau_i^{-1}\hat{z}_i^*$. Then, from the identity $\hat{z}_i^{(k+1)} = \sqrt{\tau_i \hat{z}_i^{(k)} (\kappa_i(\hat{\mathbf{z}}_i^{(k)}) - p_{\iota(i),i}\kappa_{\iota(i)}(\hat{\mathbf{z}}_{\iota(i)}^{(k)}))}$, the inequality $\hat{z}_i^{(k)} > h_i(\hat{\mathbf{z}}_{\iota(i)}^{(k)})$ is equivalent to $\hat{z}_i^{(k+1)} < \sqrt{\hat{z}_i^* \hat{z}_i^{(k)}}$ and thus we conclude that $\hat{z}_i^{(k+1)} < \sqrt{\hat{z}_i^* \hat{z}_i^{(k)}}$ for $\hat{z}_i^{(k)} \in \mathcal{U}_i^\xi$. Similar arguments show that the inequality $\hat{z}_i^{(k+1)} > \sqrt{\hat{z}_i^* \hat{z}_i^{(k)}}$ holds on the set $\mathcal{L}_i^\xi$. The desired results then follow from these relationships, thus completing the proof.

A.13 Proof of Theorem 7

It follows from Lemma 7 and Lemma 8 that, if the sequence $\{\hat{z}_i^{(k)}\}$ is a monotone sequence, then it must converge to $\hat{z}_i^*$ under our original fixed-point iterative process without modification. Suppose now that the sequence $\{\hat{z}_i^{(k)}\}$ is not a monotone sequence. Then, with the addition of the bisection step when $\hat{z}_i^{(k+1)}$ calculated from (29) lies outside of $\mathcal{I}_i^{(k)}$, the length of the interval $\mathcal{I}_i^{(k)}$ shrinks as $k \rightarrow \infty$. Moreover, the additional growth and shrinkage of the left and right endpoints of the interval, respectively, ensures that the interval length tends to shrink to 0 as $k \rightarrow \infty$. The desired result then follows from Lemma 7 and Lemma 8.

B Light-Heavy Traffic Interpolation Approximation

In this appendix, we provide a general overview of the light-heavy traffic interpolation approximation approach which includes within this context the theoretical results of Section 3.2.2. The light-heavy traffic interpolation approximation can be an effective method for analyzing otherwise intractable queueing systems. The central idea is to approximate some quantity $f(\gamma)$ of interest in the system based on a combination of n light-traffic derivatives associated with the Taylor series expansion of $f(\gamma)$ at $\gamma = 0$ and the heavy-traffic limit of $f(\gamma)$ as γ approaches its critical value μ^* . More specifically, following the approach developed by Reiman and Simon [55–57], we first scale $f(\gamma)$ by defining

$$g(\gamma) := (\mu^* - \gamma)f(\gamma), \quad \gamma \in [0, \mu^*), \quad (42)$$

and then we interpolate $g(\gamma)$ by a polynomial of order $n + 1$ of the form

$$\tilde{g}(\gamma) = a_{n+1}\gamma^{n+1} + a_n\gamma^n + \cdots + a_1\gamma + a_0, \quad (43)$$

where $a_{n+1}, a_n, \dots, a_1, a_0$ are determined from the n light-traffic derivatives associated with the Taylor series expansion of $f(\gamma)$ at $\gamma = 0$, i.e.,

$$f(\gamma) \approx f^{(0)}(0) + \gamma f^{(1)}(0) + \dots + \frac{\gamma^{n-1}}{(n-1)!} f^{(n-1)}(0) + \frac{\gamma^n}{n!} f^{(n)}(0), \quad \gamma \approx 0, \quad (44)$$

together with the heavy-traffic limit

$$\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma) f(\gamma) = h. \quad (45)$$

Here, assuming the existence of the first n derivatives of $f(\gamma)$ at $\gamma = 0$, the $f^{(k)}(0)$ in (44) denotes the order k derivative of $f(\gamma)$ at $\gamma = 0$, namely

$$f^{(k)}(0) = \left. \frac{d^k f}{d\gamma^k} \right|_{\gamma=0}, \quad k \in \{0, 1, \dots, n\}.$$

From (44) and (45), we have the matching light-traffic conditions

$$\tilde{g}^{(k)}(0) = g^{(k)}(0), \quad k \in \{0, 1, \dots, n\} \quad (46)$$

and the heavy-traffic condition

$$\tilde{g}(\mu^*) = g(\mu^*) = h. \quad (47)$$

Then, once $\tilde{g}(\gamma)$ is determined, we invert the scaling in (42) to obtain the interpolation approximation of $f(\gamma)$ as

$$\tilde{f}(\gamma) = \frac{\tilde{g}(\gamma)}{\mu^* - \gamma}, \quad \gamma \in [0, \mu^*). \quad (48)$$

The calculation of the derivatives at zero for the quantity of interest $f(\gamma)$ is clearly a focal point of the light-heavy traffic interpolation approximation approach. This is made possible within the context of Poisson (or Poisson-driven) arrival processes by the light-traffic theory rigorously developed in [57], where the authors furnish expressions that, in principle, enable us to calculate the derivative of any order n . The availability of the light traffic derivatives $f^{(k)}(0)$ for all k fully determines $f(\gamma)$ in terms of its Taylor series expansion, and thus provides a way to obtain an exact expression $\tilde{f}(\gamma) = f(\gamma)$. However, in our particular situation as well as in most practical cases, the calculations quickly become extremely involved and we therefore have to restrict ourselves to the 0th and 1st order derivatives. Nevertheless, we are strongly encouraged by the fact that our light-heavy traffic interpolation approximation for the complex unbalanced application workload with nonhomogeneous service rates $\hat{\mu}_1$ and $\hat{\mu}_2$ recovers the exact expression (3) for the expected sojourn time under the balanced workload with homogeneous service rates $\hat{\mu} = \hat{\mu}_1 = \hat{\mu}_2$ due to Nelson and Tantawi [46]. Moreover, an a posteriori comparison of the resulting approximations with simulations strongly suggests that our quadratic polynomial interpolation in γ is a very effective approximation.

Our interests are in deriving a light-heavy traffic interpolation approximation for the more difficult problem of approximating the expected sojourn time of jobs in a fork-join queueing

network comprising two parallel servers under unbalanced application workloads in which the arrivals follow a Poisson process with rate γ and the nonhomogeneous service times for each of the two servers respectively follow an exponential distribution with rates $\hat{\mu}_1$ and $\hat{\mu}_2$. The quantity of interest in the system is therefore $f(\gamma) = \mathbb{E}[T_{FJ}]$ whose approximation $\tilde{f}(\gamma) = \mathbb{E}[\tilde{T}_{FJ}]$ is based on a combination of the 0^{th} and 1^{st} order derivatives $f^{(0)}(0)$ and $f^{(1)}(0)$ associated with the Taylor series expansion of $f(\gamma)$ at $\gamma = 0$ in (44) and the heavy-traffic limit of $f(\gamma)$ as γ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$ in (45). We note that the fork-join queueing network of interest satisfies the relevant technical conditions required by the light-heavy traffic interpolation approximation approach of Reiman and Simon [55–57, 67].

Following [55–57], consider the sojourn time T_{FJ} of a tagged job that arrives at time $t = 0$ whose service times v_1 and v_2 are random variables that follow an exponential distribution with mean $\mathbb{E}[v_1] = \hat{\mu}_1$ and $\mathbb{E}[v_2] = \hat{\mu}_2$, respectively, where v_1 and v_2 are independent of each other as well as all other random variables of the fork-join queueing network. The 0^{th} order derivative can be obtained by definition as

$$f^{(0)}(0) = \mathbb{E}[\psi(\emptyset)] := \mathbb{E}[T_{FJ} \mid \text{no arrivals over } (-\infty, 0)], \quad (7)$$

noting that any arrival at time $t \in (0, \infty)$ will be serviced after the tagged job and therefore not impact T_{FJ} . We then have the following theoretical result for $f^{(0)}(0)$.

Lemma 2. *The 0^{th} order light-traffic derivative of the fork-join queueing network is given by*

$$f^{(0)}(0) = \mathbb{E}[\psi(\emptyset)] = \frac{\hat{\mu}_1^2 + \hat{\mu}_1\hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1\hat{\mu}_2(\hat{\mu}_1 + \hat{\mu}_2)}. \quad (8)$$

For the 1^{st} order derivative, consider the sojourn time T_{FJ} of a tagged job that arrives at time 0 with exponential service times v_1 and v_2 as above, given that another job arrives at time $t \in (-\infty, \infty)$ whose service times w_1 and w_2 are random variables that follow an exponential distribution with mean $\mathbb{E}[w_1] = \hat{\mu}_1$ and $\mathbb{E}[w_2] = \hat{\mu}_2$, respectively, where v_1, v_2, w_1, w_2 are independent of each other as well as all other random variables of the fork-join queueing network. The 1^{st} order derivative can be obtained by definition as

$$\mathbb{E}[\psi(\{t\})] := \mathbb{E}[T_{FJ} \mid \text{exactly one arrival at time } t \text{ over } (-\infty, \infty)], \quad (49)$$

$$f^{(1)}(0) = \int_{-\infty}^{\infty} (\mathbb{E}[\psi(\{t\})] - \mathbb{E}[\psi(\emptyset)]) dt. \quad (9)$$

We then have the following theoretical result for $f^{(1)}(0)$.

Lemma 3. *The 1^{st} order light-traffic derivative of the fork-join queueing network is given by*

$$f^{(1)}(0) = \frac{1}{\hat{\mu}_1^2} + \frac{1}{\hat{\mu}_2^2} - \frac{2}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1\hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4}. \quad (10)$$

Next, we consider the heavy-traffic limit of $f(\gamma)$ in (45) as γ approaches its critical value $\mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. Define $r := \hat{\mu}_2/\hat{\mu}_1 \in [1, \infty)$. Without loss of generality, suppose

that $\hat{\mu}_1 \leq \hat{\mu}_2$. Our approach consists of first establishing upper and lower bounds on the heavy-traffic limit h^* as $r \rightarrow 1$ and as $r \rightarrow \infty$, and then obtaining an upper-lower bound interpolation approximation for h^* . The corresponding theoretical results for the heavy-traffic limit h^* under any given $\hat{\mu}_1$ and $\hat{\mu}_2$ are presented as follows.

Lemma 4. *The heavy-traffic limit of the fork-join queueing network is bounded above and below by*

$$\lim_{\gamma \rightarrow \mu^*} (\mu^* - \gamma) \mathbb{E}[T_{FJ}] = h^* \in [1, 11/8]. \quad (50)$$

Furthermore, the upper-lower bound interpolation approximation for this heavy-traffic limit is given by

$$h^* \approx h(r) = 1 + \frac{3}{8} r^{-\beta} \in [1, 11/8], \quad r \in [1, \infty), \quad \beta \in (0, \infty). \quad (12)$$

We now exploit the above results to obtain the desired 0th and 1st order light-heavy traffic interpolation approximations $\tilde{f}_0(\gamma)$ and $\tilde{f}_1(\gamma)$ for the expected sojourn time of jobs in the two-server fork-join queueing network of interest under unbalanced application workloads.

Theorem 2. *Consider the sojourn time T_{FJ} of jobs in a fork-join queueing network comprising two parallel servers where arrivals follow a Poisson process with rate γ and service times follow an exponential distribution with nonhomogeneous rates $\hat{\mu}_1$ and $\hat{\mu}_2$ at server 1 and 2, respectively. Further let $f(\gamma) = \mathbb{E}[T_{FJ}]$ and, without loss of generality, $\hat{\mu}_1 = \mu^* := \min\{\hat{\mu}_1, \hat{\mu}_2\}$. The 0th and 1st order light-heavy traffic interpolation approximations $\mathbb{E}[T_{FJ}^{(0)}] = \tilde{f}_0(\gamma)$ and $\mathbb{E}[T_{FJ}^{(1)}] = \tilde{f}_1(\gamma)$ for $f(\gamma)$, $\gamma \in [0, \hat{\mu}_1)$, are then given by*

$$\mathbb{E}[T_{FJ}^{(0)}] = \tilde{f}_0(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \frac{h(r)}{\hat{\mu}_1} + \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)}, \quad (13)$$

$$\mathbb{E}[T_{FJ}^{(1)}] = \tilde{f}_1(\gamma) = \frac{\gamma}{\hat{\mu}_1 - \gamma} \left(\frac{1}{\hat{\mu}_1} + \frac{\hat{\mu}_1}{\hat{\mu}_2^2} - \frac{2\hat{\mu}_1}{(\hat{\mu}_1 + \hat{\mu}_2)^2} - \frac{2\hat{\mu}_1^2 \hat{\mu}_2}{(\hat{\mu}_1 + \hat{\mu}_2)^4} \right) + \frac{\hat{\mu}_1^2 + \hat{\mu}_1 \hat{\mu}_2 + \hat{\mu}_2^2}{\hat{\mu}_1 \hat{\mu}_2 (\hat{\mu}_1 + \hat{\mu}_2)}. \quad (14)$$

Remark 2. *It is important to note that, upon substituting $\hat{\mu}_1 = \hat{\mu}_2 = \hat{\mu}$ in (13) and (14), our light-heavy traffic interpolation approximations recover the exact expected sojourn time of jobs in the two-server fork-join queueing network of interest under balanced application workloads, as specifically expressed in (3).*

C Mathematical Analysis of Reserved Allocation

In this appendix, we provide some additional details on the matrix-analytic analysis of the QCS environment under the reservation-based resource allocation policy in Section 4.

The matrices corresponding to the non-homogeneous boundary of the state space $\hat{\Omega}$ are

given by

$$\mathbf{B}_{00} = [-\lambda], \quad \mathbf{B}_{01} = [\mathbf{B}_{01}^P \quad \mathbf{B}_{01}^S], \quad \mathbf{B}_{10} = [\mathbf{B}_{10}^P \quad \mathbf{B}_{10}^S]^\top, \quad (51)$$

$$\mathbf{B}_{01}^P = [p_{\parallel}\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad \mathbf{B}_{01}^S = [(1-p_{\parallel})\lambda \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0], \quad (52)$$

$$\mathbf{B}_{10}^P = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^P], \quad \mathbf{B}_{10}^S = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \mu_T^S], \quad (53)$$

$$\mathbf{B}_{11} = \begin{bmatrix} \mathbf{B}_{11}^P & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{11}^S \end{bmatrix}, \quad \mathbf{B}_{11}^P = \begin{bmatrix} \mathbf{B}_{11}^{P_{pp}} & \mathbf{B}_{11}^{P_{ps}} \\ \mathbf{0} & \mathbf{B}_{11}^{P_{ss}} \end{bmatrix}, \quad \mathbf{B}_{11}^S = \begin{bmatrix} \mathbf{B}_{11}^{S_{pp}} & \mathbf{B}_{11}^{S_{ps}} \\ \mathbf{B}_{11}^{S_{sp}} & \mathbf{B}_{11}^{S_{ss}} \end{bmatrix}, \quad (54)$$

$$\mathbf{B}_{11}^{P_{pp}} = \begin{bmatrix} (\lambda, \mu_I^P) & 0 & 0 & \mu_I^P \\ 0 & (\lambda, \mu_G^{PP}) & 0 & 0 \\ 0 & 0 & (\lambda, \mu_Q^{PP}) & 0 \\ 0 & \mu_Q^{PP} & \mu_G^{PP} & (\lambda, \mu_Q^{PP}, \mu_G^{PP}) \end{bmatrix}, \quad (55)$$

$$\mathbf{B}_{11}^{P_{ps}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ (1-p_{\circ})\mu_G^{PP} & 0 & 0 & p_{\circ}\mu_G^{PP} & 0 \\ (1-p_{\circ})\mu_Q^{PP} & 0 & 0 & p_{\circ}\mu_Q^{PP} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (56)$$

$$\mathbf{B}_{11}^{P_{ss}} = \begin{bmatrix} (\lambda, \mu_G^{PS}) & \mu_G^{PS} & 0 & 0 & 0 \\ 0 & (\lambda, \mu_Q^{PS}) & 0 & 0 & \mu_Q^{PS} \\ 0 & 0 & (\lambda, \mu_G^{PS}) & 0 & \mu_G^{PS} \\ 0 & 0 & \mu_Q^{PS} & (\lambda, \mu_Q^{PS}) & 0 \\ 0 & 0 & 0 & 0 & (\lambda, \mu_T^P) \end{bmatrix}, \quad (57)$$

$$\mathbf{B}_{11}^{S_{pp}} = \begin{bmatrix} (\lambda, \mu_I^S) & 0 & 0 & 0 \\ 0 & (\lambda, \mu_G^{SP}) & 0 & 0 \\ 0 & 0 & (\lambda, \mu_Q^{SP}) & 0 \\ 0 & \mu_Q^{SP} & \mu_G^{SP} & (\lambda, \mu_G^{SP}, \mu_Q^{SP}) \end{bmatrix}, \quad (58)$$

$$\mathbf{B}_{11}^{S_{ps}} = \begin{bmatrix} (1-p_{\circ})\mu_I^S & 0 & 0 & p_{\circ}\mu_I^S & 0 \\ 0 & 0 & 0 & 0 & \mu_G^{SP} \\ 0 & 0 & 0 & 0 & \mu_Q^{SP} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (59)$$

$$\mathbf{B}_{11}^{S_{sp}} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu_Q^{SS} \\ 0 & 0 & 0 & \mu_G^{SS} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}_{11}^{S_{ss}} = \begin{bmatrix} (\lambda, \mu_G^{SS}) & \mu_G^{SS} & 0 & 0 & 0 \\ 0 & (\lambda, \mu_Q^{SS}) & 0 & 0 & 0 \\ 0 & 0 & (\lambda, \mu_G^{SS}) & 0 & 0 \\ 0 & 0 & \mu_Q^{SS} & (\lambda, \mu_Q^{SS}) & 0 \\ 0 & 0 & 0 & 0 & (\lambda, \mu_T^S) \end{bmatrix}, \quad (60)$$

where $(x, y) = -(x + y)$ and $(x, y, z) = -(x + y + z)$. Here, the matrix $\mathbf{B}_{10}$ defines the rate of transitions from states in $\hat{\Omega}_1$ to the state $\hat{\Omega}_0$, the matrix $\mathbf{B}_{01}$ defines the rate of transitions from the state $\hat{\Omega}_0$ to states in $\hat{\Omega}_1$, and the offdiagonal elements of the matrix $\mathbf{B}_{11}$ define the rate of transitions among states in $\hat{\Omega}_1$. Similarly, for the homogeneous portion of the state space $\hat{\Omega}$, the matrix $\mathbf{A}_{-1}$ defines the rate of transitions from states in $\hat{\Omega}_{i+1}$ to states

in $\hat{\Omega}_i$, the matrix $\mathbf{A}_1$ defines the rate of transitions from states in $\hat{\Omega}_i$ to states in $\hat{\Omega}_{i+1}$, and the offdiagonal elements of the matrix $\mathbf{A}_0$ define the rate of transitions among states in $\hat{\Omega}_i$, $i \in \mathbb{Z}_{\geq 1}$. Hence, the matrices for the homogeneous portion of the state space are given by $\mathbf{A}_1 = \lambda \mathbf{I}$, $\mathbf{A}_0 = \mathbf{B}_{11}$, and

$$\mathbf{A}_{-1} = \begin{bmatrix} \mathbf{A}_{-1}^{PP} & \mathbf{A}_{-1}^{PS} \\ \mathbf{A}_{-1}^{SP} & \mathbf{A}_{-1}^{SS} \end{bmatrix}, \quad (61)$$

$$\mathbf{A}_{-1}^{PP} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{\parallel} \mu_T^P & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{-1}^{PS} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (1-p_{\parallel})\mu_T^P & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (62)$$

$$\mathbf{A}_{-1}^{SP} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{\parallel} \mu_T^S & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{A}_{-1}^{SS} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (1-p_{\parallel})\mu_T^S & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (63)$$

where $\mathbf{I}$ denotes the identity matrix of appropriate dimension.

The stationary distribution $\boldsymbol{\pi}$ can be used to obtain performance measures and quantities of interest beyond those given at the end of Section 4. This includes the asymptotic decay rate of the tail distribution of the queue length process as another measure of interest. Let η denote this decay rate, which is also called the caudal characteristic [50, 51]. Due to the irreducibility of the Markov process $\{X(t); t \in \mathbb{R}_+\}$ and the irreducibility of the generator matrix $\mathbf{A}$, we either have $\eta = \text{sp}(\mathbf{R})$ when the matrix $\mathbf{R}$ is irreducible, or otherwise have $\eta = \text{sp}(\mathbf{R}_{11})$ where, after appropriate row and column permutations,

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{R}_{22} \end{bmatrix}$$

with $\mathbf{R}_{11}$ irreducible and $\mathbf{R}_{22}$ strictly upper triangular. We then conclude that $\mathbf{R}^z = O(\eta^z)$, as $z \rightarrow \infty$. Let $\mathbf{u}$ and $\mathbf{v}$ be the left and right eigenvectors corresponding to η that are strictly positive and normalized by $\mathbf{u}\mathbf{e} = 1$ and $\mathbf{u}\mathbf{v} = 1$. From the properties of the Markov process

$\{X(t); t \in \mathbb{R}_+\}$, we know that [61]

$$\mathbf{R}^z = \eta^z \mathbf{v} \cdot \mathbf{u} + o(\eta^z), \quad \text{as } z \rightarrow \infty,$$

which together with the quadratic matrix equation in Theorem 3 yields

$$\boldsymbol{\pi}_{z+1} \mathbf{e} = \boldsymbol{\pi}_1 \mathbf{v} \eta^z + o(\eta^z), \quad \text{as } z \rightarrow \infty. \quad (64)$$

It then follows that

$$\mathbb{P}[Q \geq z+1] = \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1-\eta} \eta^z + o(\eta^z), \quad \text{as } z \rightarrow \infty, \quad (65)$$

and thus

$$\lim_{z \rightarrow \infty} \frac{\mathbb{P}[Q \geq z+1]}{\eta^z} = \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1-\eta}, \quad (66)$$

or equivalently

$$\mathbb{P}[Q \geq z+1] \sim \frac{\boldsymbol{\pi}_1 \mathbf{v}}{1-\eta} \eta^z, \quad \text{as } z \rightarrow \infty, \quad (67)$$

where $f(z) \sim g(z)$ denotes that $\lim_{z \rightarrow \infty} f(z)/g(z) = 1$. Hence, in addition to providing the stability criterion for the QCSC environment, η is indicative of the tail behavior of the stationary queue length distribution.

Note that the caudal characteristic can be obtained without having to first solve for the matrix $\mathbf{R}$ in Theorem 3. Define the matrix $\tilde{\mathbf{A}}(s) := \mathbf{A}_1 + s\mathbf{A}_0 + s^2\mathbf{A}_{-1}$, for $0 < s \leq 1$. When the generator matrix $\mathbf{A}$ is irreducible, this matrix $\tilde{\mathbf{A}}(s)$ is irreducible with nonnegative off-diagonal elements. Let $\tilde{\eta}(s)$ denote the spectral radius of the matrix $\tilde{\mathbf{A}}(s)$. Then, under the above assumptions, η is the unique solution in $(0,1)$ of the equation $\tilde{\eta}(s) = 0$. This solution can be directly computed, although more efficient methods have been developed [8].

Another set of performance metrics of interest concern the stationary utilization of the system CPUs, QPUs and GPUs composing the QCSC environment. For each level i , $i \in \mathbb{Z}_+$, the states with elements $k \in \{I, T\}$ represent execution on the system CPUs, the states with elements $q = 1$ represent execution on the system QPUs, and the states with elements $g = 1$ represent execution on the system GPUs. The corresponding stationary utilizations can then be expressed as

$$U_C = \sum_{n=0}^{\infty} \pi_{n+1} \nu_C = \boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_C, \quad (68)$$

$$U_Q = \sum_{n=0}^{\infty} \pi_{n+1} \nu_Q = \boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_Q, \quad (69)$$

$$U_G = \sum_{n=0}^{\infty} \pi_{n+1} \nu_G = \boldsymbol{\pi}_1 (\mathbf{I} - \mathbf{R})^{-1} \nu_G, \quad (70)$$

respectively, where the ℓ^{th} element of the vector ν_C contains a 1 if $k \in \{I, T\}$ in the corresponding state of the homogeneous portion and contains a 0 otherwise, the ℓ^{th} element of the vector ν_Q contains a 1 if $q = 1$ in the corresponding state of the homogeneous portion and contains a 0 otherwise, and the ℓ^{th} element of the vector ν_G contains a 1 if $g = 1$ in the corresponding state of the homogeneous portion and contains a 0 otherwise.

D Numerical Experiments

In this appendix we present additional empirical results from our numerical experiments in Section 6. Specifically, Figures 6 through 8 provide expanded versions of the numerical results in Section 6 for $\mathcal{C} = 21$. Then, Figures 9 through 11 provide versions of the numerical results in Section 6 for $\mathcal{C} = 42$.

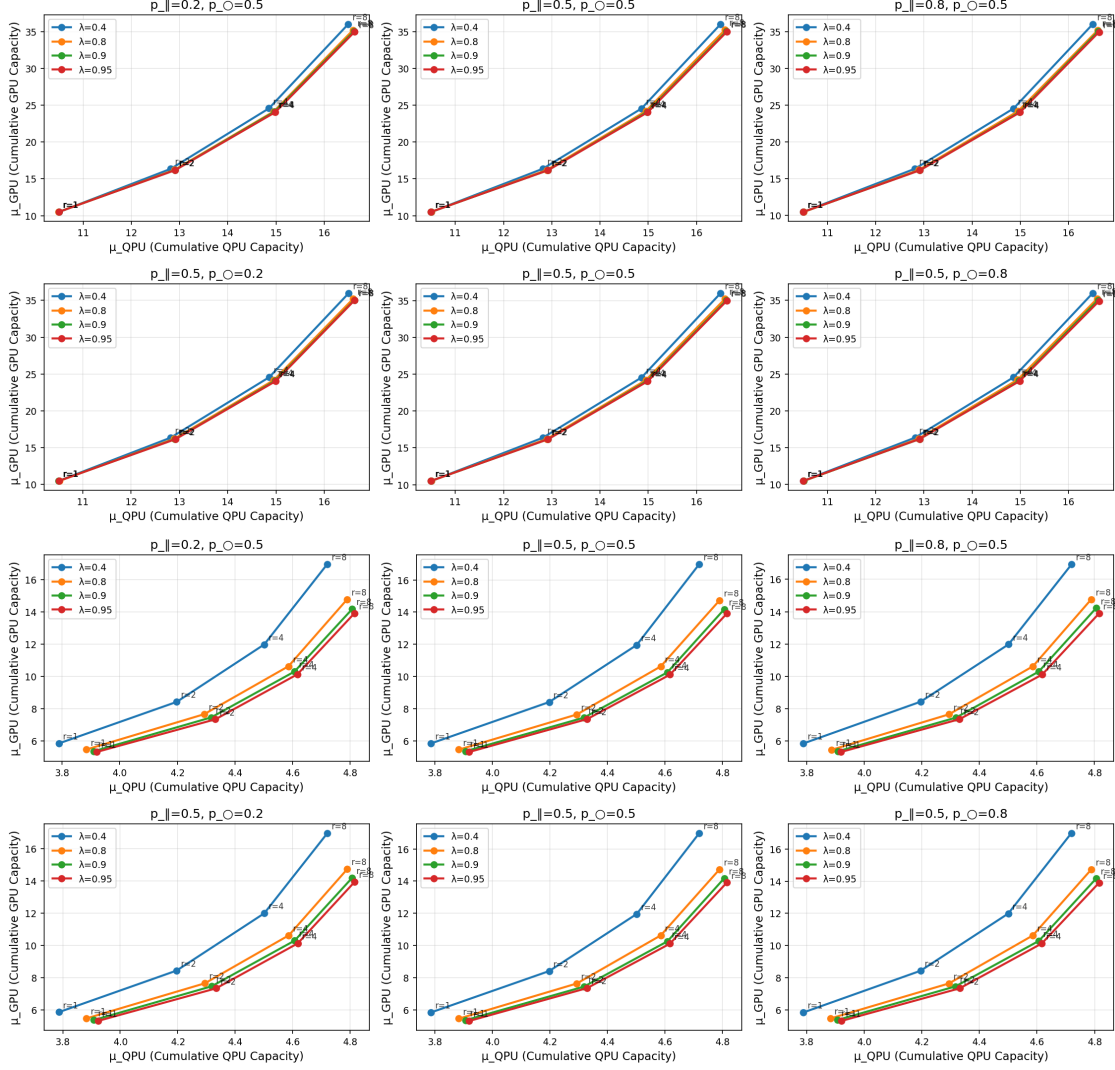


Figure 6: The optimal resource capacity allocations for system QPUs and GPUs under different parameters of our QCSC models. The y-axis depicts the cumulative GPU capacities and the x-axis depicts the cumulative QPU capacities for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$ and $r \in \{1, 2, 4, 8\}$, under unit costs for QPUs and GPUs in the top two rows and under costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom two rows. The first and third rows fix $p_\circ = 0.5$ while varying $p_\parallel \in \{0.2, 0.5, 0.8\}$, whereas the second and fourth rows fix $p_\parallel = 0.5$ while varying $p_\circ \in \{0.2, 0.5, 0.8\}$.

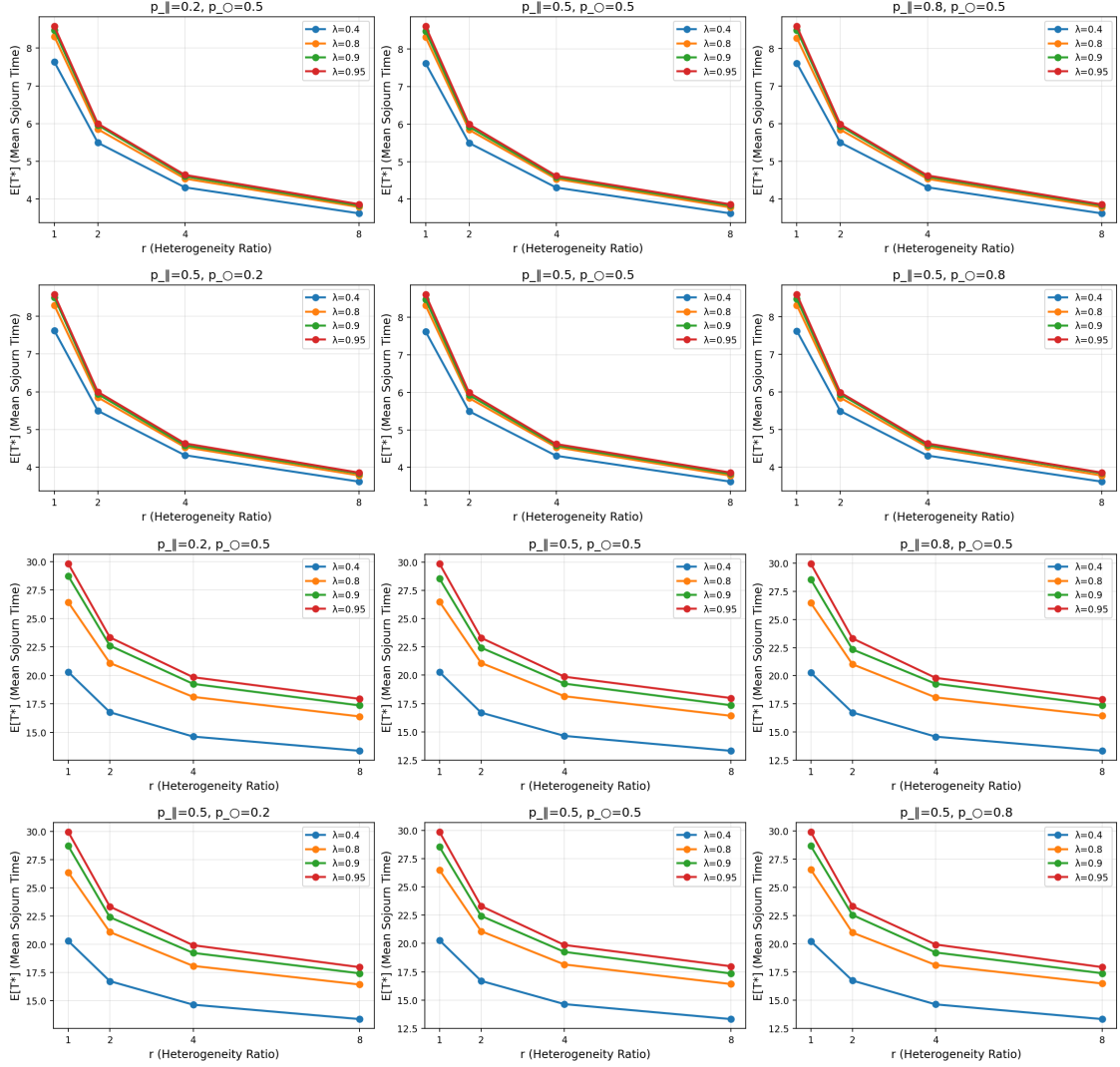


Figure 7: The expected sojourn time under the optimal resource capacity allocations as a function of the heterogeneity ratio $r \in \{1, 2, 4, 8\}$ for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$, with unit costs for QPUs and GPUs in the top two rows and with costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom two rows. The first and third rows fix $p_o = 0.5$ while varying $p_{\parallel} \in \{0.2, 0.5, 0.8\}$, whereas the second and fourth rows fix $p_{\parallel} = 0.5$ while varying $p_o \in \{0.2, 0.5, 0.8\}$.

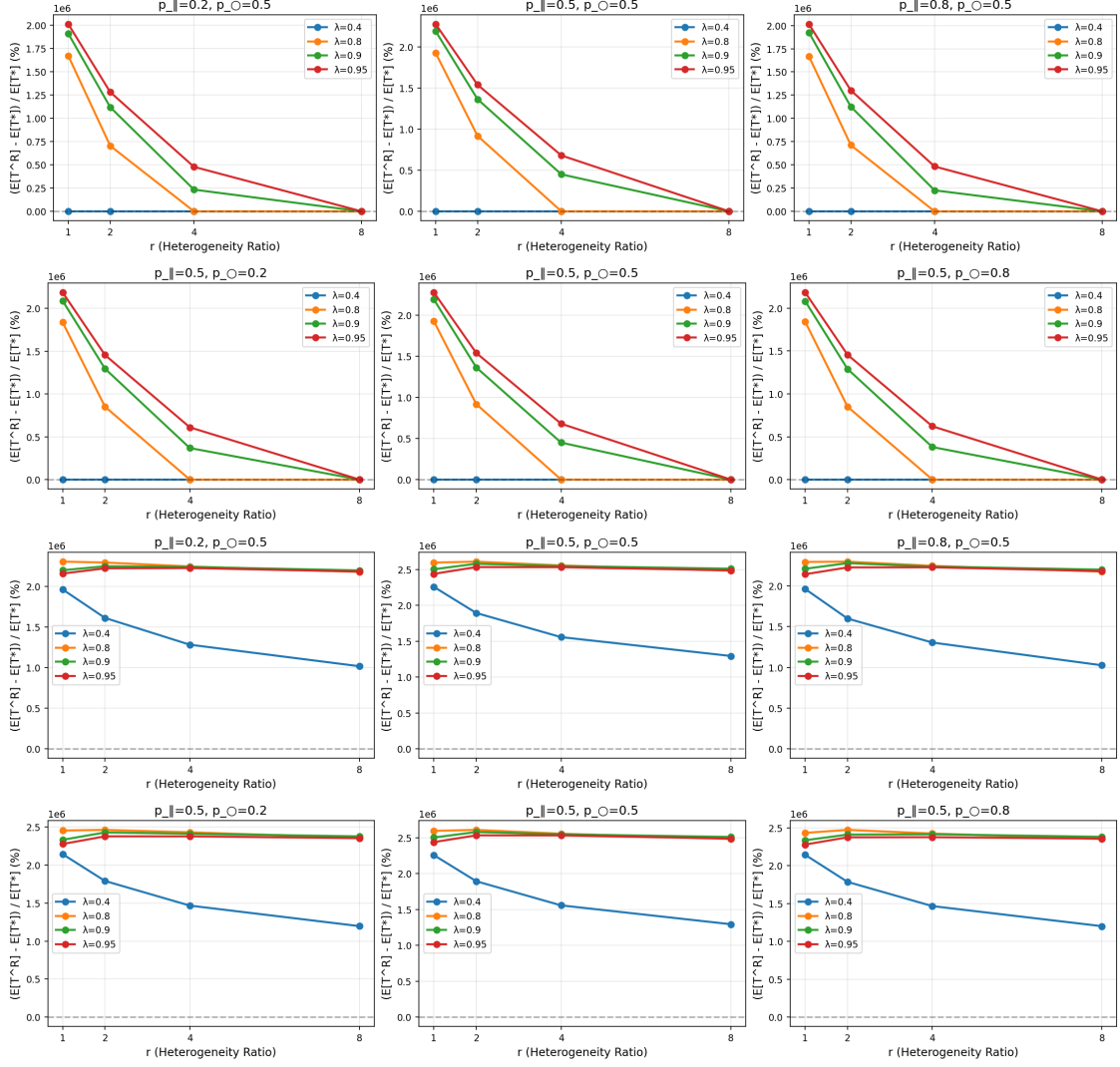


Figure 8: The relative improvement in expected sojourn time under the optimal resource capacity allocations over that under the reservation-based resource allocation policy as a function of the heterogeneity ratio $r \in \{1, 2, 4, 8\}$ for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$, with unit costs for QPUs and GPUs in the top two rows and with costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom two rows. The first and third rows fix $p_o = 0.5$ while varying $p_{||} \in \{0.2, 0.5, 0.8\}$, whereas the the second and fourth rows fix $p_{||} = 0.5$ while varying $p_o \in \{0.2, 0.5, 0.8\}$.

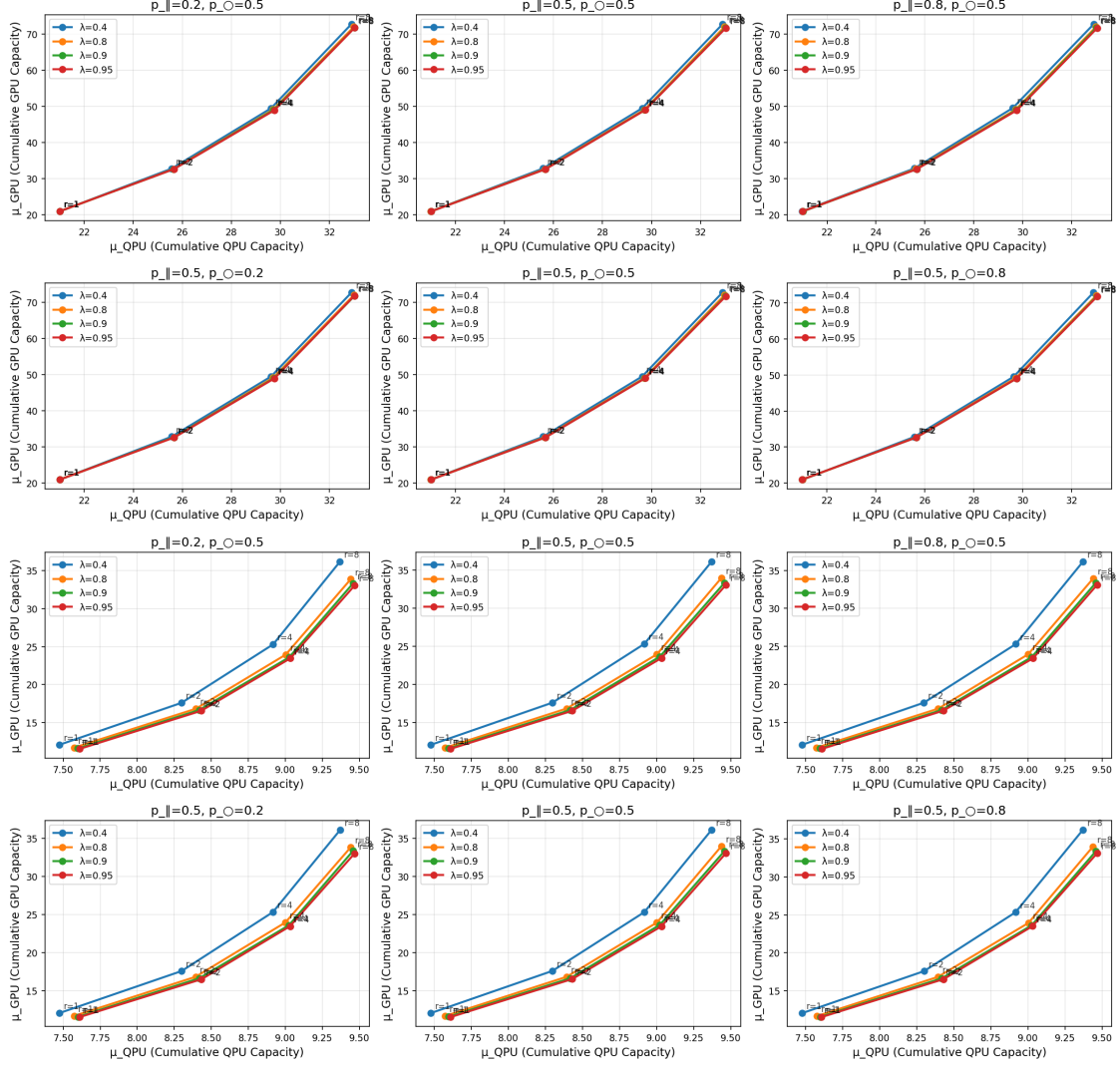


Figure 9: The optimal resource capacity allocations for system QPUs and GPUs under different parameters of our QCSC models. The y-axis depicts the cumulative GPU capacities and the x-axis depicts the cumulative QPU capacities for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$ and $r \in \{1, 2, 4, 8\}$, under unit costs for QPUs and GPUs in the top two rows and under costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom two rows. The first and third rows fix $p_{\perp} = 0.5$ while varying $p_{||} \in \{0.2, 0.5, 0.8\}$, whereas the second and fourth rows fix $p_{||} = 0.5$ while varying $p_{\perp} \in \{0.2, 0.5, 0.8\}$.

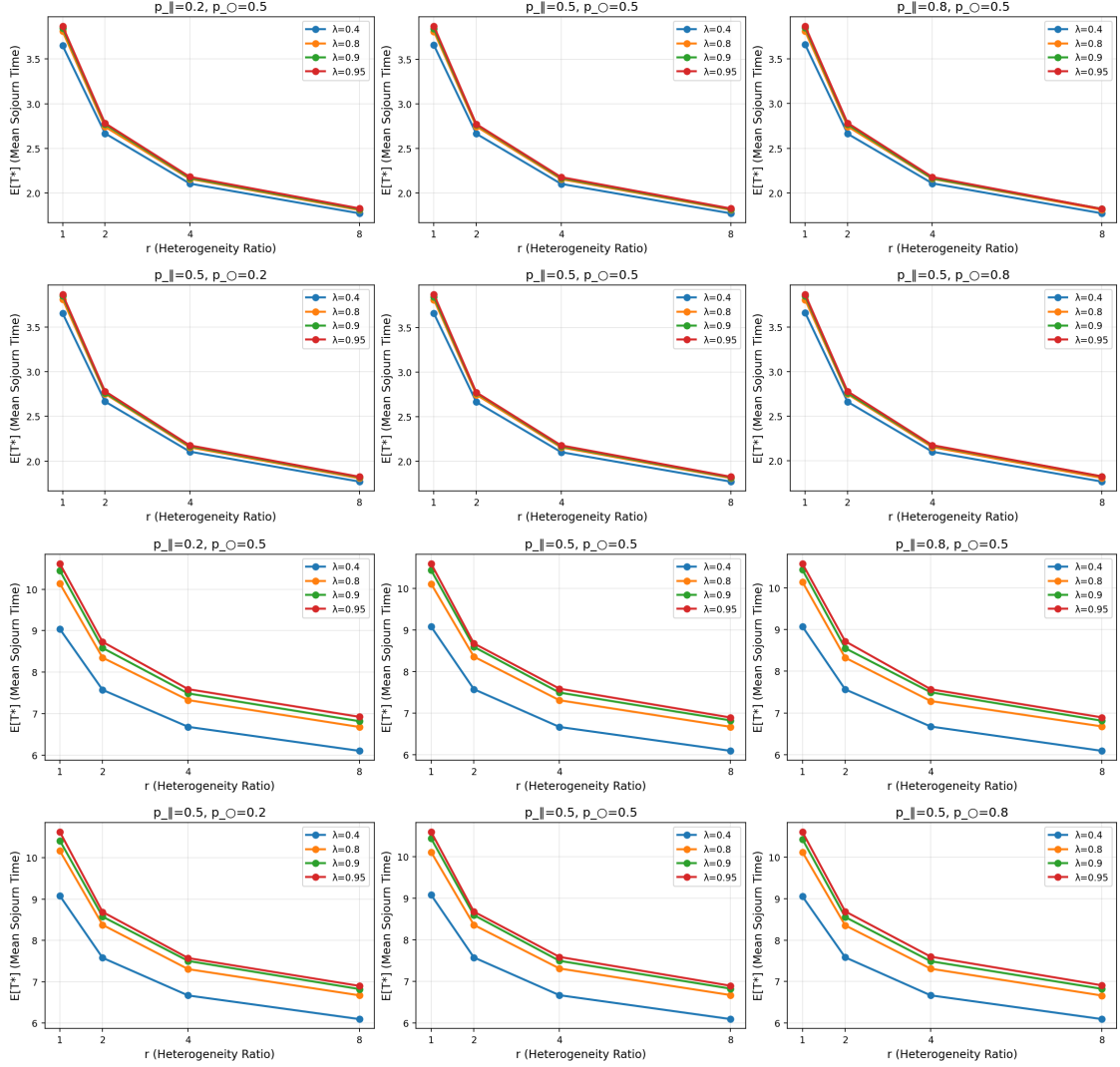


Figure 10: The expected sojourn time under the optimal resource capacity allocations as a function of the heterogeneity ratio $r \in \{1, 2, 4, 8\}$ for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$, with unit costs for QPUs and GPUs in the top two rows and with costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom two rows. The first and third rows fix $p_{\ominus} = 0.5$ while varying $p_{||} \in \{0.2, 0.5, 0.8\}$, whereas the second and fourth rows fix $p_{||} = 0.5$ while varying $p_{\ominus} \in \{0.2, 0.5, 0.8\}$.

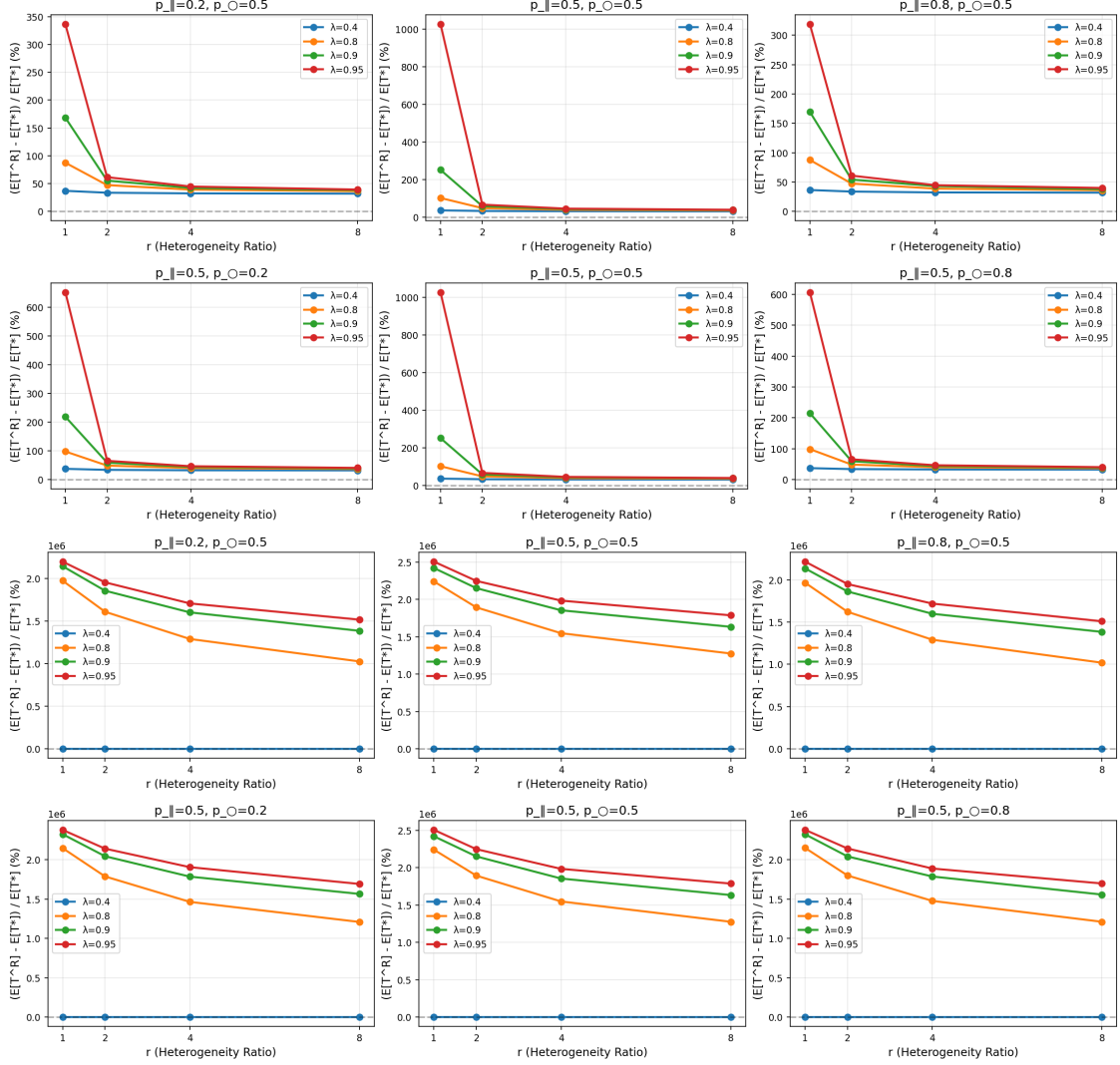


Figure 11: The relative improvement in expected sojourn time under the optimal resource capacity allocations over that under the reservation-based resource allocation policy as a function of the heterogeneity ratio $r \in \{1, 2, 4, 8\}$ for different $\lambda \in \{0.4, 0.8, 0.9, 0.95\}$, with unit costs for QPUs and GPUs in the top two rows and with costs $c_i = 4$ for QPUs and costs $c_i = 1$ for GPUs in the bottom two rows. The first and third rows fix $p_{\perp} = 0.5$ while varying $p_{\parallel} \in \{0.2, 0.5, 0.8\}$, whereas the the second and fourth rows fix $p_{\parallel} = 0.5$ while varying $p_{\perp} \in \{0.2, 0.5, 0.8\}$.